Fertility-Based Discrimination*

Andrea Di Giovan Paolo† Giacomo Marcolin‡

Job Market Paper

November 9, 2026

[Click here for the most recent version]

Abstract

We study employer discrimination based on expectations about women's future fertility. The identification challenge is that expectations about fertility cannot be varied independently of the characteristics from which employers infer them. We show that discrimination can be identified from variation in employers' cost of future fertility, without requiring variation in fertility expectations. We exploit the Pregnancy Discrimination Act of 1978, which first provided federal employment protection to pregnant workers, raising firms' expected costs of employing women more likely to become pregnant. Using a difference-in-differences design comparing newly affected states with states that already had comparable protections, we find that the PDA reduced employment among women of childbearing age by 6.6%. The decline is concentrated among women with higher predicted future fertility, while employment of men and older women is largely unaffected. Our estimates imply that expected-fertility discrimination accounts for at least 27% of the observed gender employment gap.

*We are deeply grateful to Abi Adams, Nicola Bianchi, Elisa Jacome, Joel Mokyr, Alessandro Pavan, Nicola Persico, and Silvia Vannutelli for their guidance and advice on this project. For very helpful comments, we thank Martha Bailey, Devis Decet, Claudia Goldin, Jonathan Gruber, Walker Hanlon, Kevin Lang, Camille Landais, Hamish Low, Conrad Miller, Claudia Olivetti, Myera Rashid, Maddalena Ronchi, Matteo Ruzzante, Molly Schnell, Almudena Sevilla, Garima Sharma, Brenden Timpe, Melanie Wasserman, and seminar and conference participants at Northwestern University, LSE, NBER GITE Summer Institute 2025, SOLE 2026, EEA 2026, Marco Fanno Workshop, IZA PhD Workshop, IEA-WB. Funding from the Morton Schapiro Center for Applied Economics and the Institute for Humane Studies is gratefully acknowledged.

†Department of Economics, Northwestern University. Email: andreadigiovanpaolo@u.northwestern.edu
‡Department of Economics, Northwestern University. Email: marcolin.giacomo@u.northwestern.edu

1 Introduction

Fertility plays a central role in gender disparities in the labor market. After childbirth, men's and women's careers often diverge sharply, a pattern known as "child penalties" (e.g., Goldin, 1990; Kleven et al., 2024). Yet substantial gender gaps are present even before and in the absence of children (Gallen, 2023; Folke and Rickne, 2026; Landais et al., 2026). Fertility may contribute to these gaps if employers' decisions are shaped by beliefs about workers' *future* fertility. Childbirth can impose costs on firms through temporary replacement, accommodations, benefits, or productivity disruptions. Employers may anticipate these costs and treat women differently based on whether they expect them to have children. We refer to this behavior as "expected-fertility discrimination". Measuring its importance is essential for understanding gender disparities and how to address them, improving fairness and the efficient allocation of talent (Hsieh et al., 2019).

Proving that employers act on fertility expectations is particularly challenging: expectations cannot be varied while holding everything else fixed. Standard approaches to measure disparate treatment vary the characteristic on which it may be based while holding other factors fixed (e.g., Bertrand and Mullainathan, 2004; Arnold et al., 2018). When that characteristic is an employer belief, varying it requires changing the information on which the belief is based. That information may itself affect employers' decisions through other channels. Age, for example, is informative about future fertility, but differential treatment by age could also reflect beliefs about experience or productivity. Consequently, variation in fertility expectations generally cannot identify expected-fertility discrimination, let alone quantify its contribution to observed gender disparities.

In this paper, we identify and quantify expected-fertility discrimination by exploiting a plausibly exogenous change in the cost of future pregnancy to employers. Our approach relies on a change in the consequences of pregnancy rather than in the worker characteristics from which employers infer its likelihood. We formally show that this approach allows us to isolate differential treatment based on fertility expectations. We implement this strategy exploiting the U.S. Pregnancy Discrimination Act of 1978 (PDA). The PDA was a landmark anti-discrimination policy that increased the cost to firms of dismissing workers once they became pregnant. It thereby raised the expected cost of employing women whom firms perceived as more likely to become pregnant. If employers act on fertility expectations, the reform should reduce employment or wages more among women with higher expected fertility. We use the response to the reform to identify expected-fertility discrimination and provide, to our knowledge, the first estimate of its contribution to observed gender gaps.

We first develop a conceptual framework to formalize the identification problem and

our solution. We begin with a standard hiring problem: an employer hires a worker if the value of her marginal product exceeds her expected employment cost. If the employer anticipates future fertility costs and acts on fertility expectations, this cost consists of the wage plus fertility costs weighted by the perceived probability of future pregnancy. Worker characteristics determine this probability but may also affect productivity and wages. Hence, differences in hiring decisions across workers with different expected fertility do not, by themselves, establish that employers act on fertility expectations. Variation in fertility costs provides a solution: if expected-fertility discrimination is present, workers with higher expected fertility will be less likely to be hired when costs increases.

In practice, we exploit policy-induced changes in fertility costs and observe their effects on employment and wages in equilibrium, rather than on individual hiring decisions. We therefore embed the hiring problem in a competitive labor-market model, following Card and Lemieux (2001), to characterize how the employment and wages of both women and men respond to higher fertility costs. In this standard framework, if expected-fertility discrimination is present, higher costs reduce fertile-age women's employment and wages, while spillovers to other workers are *ex ante* ambiguous.

The PDA increased firms' fertility costs by introducing federal protections for pregnant workers. Although the Equal Pay Act of 1963 restricted sex-based pay differences for equal work and Title VII of the Civil Rights Act of 1964 prohibited sex discrimination more broadly (Bailey, Helgerman, et al., 2024), pregnancy remained effectively excluded from these protections (Wolkinson and Liberson, 1982). The PDA changed this in 1978 by requiring pregnant workers to be treated the same as workers with comparable temporary disabilities. A central implication was employment protection: dismissing a worker because of pregnancy now exposed firms to legal costs. For firms that would otherwise dismiss pregnant workers, the Act therefore raised the cost of future pregnancies through either expected sanctions or the costs of retaining the worker.

To identify the causal effect of the PDA, we exploit plausibly exogenous variation generated by states that adopted comparable policies prior to the federal reform. Employers in these states should have experienced minimal changes in costs of future pregnancies as a result of the PDA. Moreover, at the time, women's rights were a bipartisan issue supported in both states that had and had not adopted PDA-like protections prior to 1978. Hence, early adoption did not reflect distinctively favorable attitudes towards these issues, but rather the institutional and political conditions that allowed this support to translate into policy (Ramanathan, 2021). Empirically, we document that women's employment was trending similarly in states with and without pregnancy protections prior to 1978.

Our empirical strategy is a difference-in-differences design that compares changes around

1978 in states first subject to pregnancy protections through the PDA, which we call "PDA-treated" states, with changes in states where comparable protections were already in place, which we call "PDA-control" states. Identification relies on a parallel trends assumption between outcomes of PDA-treated states, had they remained untreated, and of PDA-control states, which are treated throughout. As such, we must assume that the effects of early state policies in PDA-control states are constant, with no dynamic component. We restrict the sample to states that are treated sufficiently before 1978, whose effects are more likely to have stabilized, and assess the plausibility of this assumption looking at trends prior to 1978.

We use several complementary sources of individual-level data. The Current Population Survey Annual Social and Economic Supplement (CPS ASEC) provides large repeated cross-sections with information on annual employment, wages, and fertility; the basic monthly CPS provides higher-frequency measures of current employment and employment transitions; and the Panel Study of Income Dynamics (PSID) allows us to follow individuals longitudinally, measure transitions into employment directly, and observe completed fertility.

We first show that employment among women of childbearing age decreased by 4.5 percentage points as a consequence of the PDA. This corresponds to a decline of approximately 6.6% relative to pre-PDA employment levels. We find no evidence of differential trends in this outcome prior to 1978. Then, compared to PDA-control states, fertile-age women's employment in PDA-treated states declines within a year of the PDA and remains lower thereafter. The decline is concentrated at ages at which the observed fertility rate is higher and progressively disappears as women move beyond childbearing age. Employment of women past fertility age is largely unaffected. Additionally, we use data on completed fertility to predict which women were more likely to have children in the near future based on their characteristics and confirm that the employment decline is concentrated among women with high predicted fertility. By contrast, we do not observe a similar relationship between expected fertility, as measured by age, and employment declines for men. Taken together with the causal response to the PDA, these patterns establish that employers discriminate against women based on expected fertility.

Adjustment through observed wages is considerably smaller. We estimate a negative, but imprecise, change in hourly wages. Its interpretation is complicated by the fact that wages are observed only for employed workers and the PDA itself changes selection into employment. However, relatively small wage responses may partly reflect the high elasticity of female labor supply during this period (Elder et al., 2025) and may also reflect existing wage rigidities.

While men's employment is mostly unaffected, we observe a decline in their wages. This response shows that, in equilibrium, discrimination can negatively affect men as well. Through the lenses of our framework, this happens when male and female effective labor are Edgeworth

complements in production. Patterns across occupations support this explanation, with declines for women's employment and men's wages being concentrated in different parts of the skill distribution. For women of childbearing age in clerical occupations, such as secretarial and other office jobs, and operative occupations, primarily in manufacturing. For men, particularly young men, in professional, managerial and skilled-trade and supervisory occupations. Taken together, the responses across demographic groups imply a decline in aggregate employment. We corroborate this conclusion using administrative data from County Business Patterns and local GDP estimates: employment-to-population rates fall following the PDA, establishments become smaller on average, and per-capita GDP declines by approximately 2.5%.

Hiring is an important margin through which the PDA reduced fertile-age women's employment. Using panel data, we document a large, but imprecise, decline in transitions into employment. This anticipatory response affecting women *at risk* of pregnancy is induced by stronger employment protections for women who *become* pregnant. Ideally, we would compare the decline in hiring with the reduction in involuntary separations among pregnant women, but data limitations prevent us from estimating the latter effect. Instead, we provide historical evidence that dismissing women upon pregnancy was a common practice before the PDA. Using post-reform survey data, we show that involuntary separations among pregnant women remained prevalent, suggesting that the reform did not eliminate this practice. As for fertility itself, we find no effect on aggregate, but a decline in fertility conditional on employment, consistent with employers screening out women with higher expected fertility.

Finally, we use the estimated effects of the PDA to quantify that expected-fertility discrimination explains a large share of the gender employment gap. We focus on employment because it is strongly affected by the policy and, unlike wages, it is not affected by changes in selection. Among individuals ages 18–35, the post-PDA gender employment gap in our setting is approximately 17.4 p.p. Had the PDA not been enacted, this gap would have been 4.7 p.p. smaller. Thus, the PDA-induced increase in expected-fertility discrimination alone accounts for approximately 27% of the observed gender employment gap.

We interpret this estimate as a lower bound on the overall contribution of expected-fertility discrimination. Eliminating the PDA does not eliminate the pregnancy-related costs that firms would face even in its absence. Pregnancy can still generate replacement costs, work reorganization, accommodation costs, and productivity disruptions, so our quasi-experiment isolates only the additional discrimination generated by the PDA-induced increased exposure to these costs. The 27% estimate is therefore a lower bound if eliminating these baseline costs would further reduce the gender employment gap. This condition is not mechanical, however, because changes in fertility costs can affect men's employment through general-

equilibrium spillovers. We therefore use our theoretical framework and empirically discipline its parameters to verify that this condition holds across plausible parameterizations. We conclude that expected-fertility discrimination explains *at least* 27% of the gender employment gap among fertile-age individuals. Because expected-fertility discrimination is itself a form of gender discrimination, this is a lower bound also on the contribution of gender discrimination.

Related literature. Our paper advances the understanding of the sources of gender disparities in the labor market by measuring the importance of discrimination based on expected fertility. We build on a large literature studying gender disparities, discrimination, and anti-discrimination policies (Zabalza and Tzannatos, 1985; Goldin and Rouse, 2000; Azmat and Petrongolo, 2014; Neumark and Stock, 2006; Thomas, 2020; Bailey, Helgerman, et al., 2024; Maloney and Neumark, 2025; Gentile Passaro et al., 2026); see Blau and Kahn (2017) and Goldin (2023) for reviews. A growing body of evidence suggests that employers act on women's expected fertility (Becker et al., 2019; Bamieh and Ziegler, 2023; Gallen, 2023; Agarwal et al., 2024; Sun, 2025; Li and Xiao, 2025; Landais et al., 2026). Yet causal and quantitative evidence remains limited because identification requires isolating the expected-fertility channel from other forces affecting employers' decisions. He et al. (2023) exploit a policy change in China that made only-child status an excludable fertility predictor, documenting the presence of expected-fertility discrimination through a correspondence study. We instead exploit a change in the cost of future pregnancy while leaving employers' information about workers unchanged. Our approach has two main advantages. First, it captures discrimination arising from the full set of characteristics employers use to predict fertility, not just from excludable predictors. This matters especially for quantification: most predictors of future fertility are not excludable, including age, which we show to be a major driver of differential treatment. Second, studying expected-fertility discrimination in the labor market allows us to measure its presence and consequences in equilibrium. Including spillovers to men and the contribution to aggregate gender disparities.

Our paper contributes to the broader literature on discrimination in three ways. First, our framework formalizes an identification problem arising when disparate treatment is based on employer beliefs rather than on a single observable characteristic. Rose (2023) makes a similar point for race when it is viewed as a construct inferred from multiple signals and discusses identification through variation in those signals. We extend this problem to beliefs about latent attributes and show that identification can also be achieved by varying the consequences of attributes for the employer's payoff.Our framework and proposed solution apply beyond expected fertility to settings in which employers act on beliefs about unobserved characteristics, such as past criminal conduct or drug use, studied in the case

of race discrimination (Wozniak, 2015; Agan and Starr, 2018), or future events, such as property damage (Ewens et al., 2014; Arefeva et al., 2024) and loan repayment (Dobbie et al., 2021; Brock and De Haas, 2023). Second, while extensive evidence documents the existence of discrimination, particularly by race and gender, much less is known about how much it contributes to observed disparities (Lang and Kahn-Lang Spitzer, 2020). We show that expected-fertility discrimination, and thus also gender discrimination, accounts for a sizeable share of the gender employment gap. Third, we document that discrimination can harm workers beyond the directly discriminated group and reduce aggregate outcomes. Prior work emphasizes how discrimination can distort allocation and investments over time (Hsieh et al., 2019), we show that sizeable aggregate costs can arise contemporaneously, among the same cohorts exposed to an increase in discrimination.

Finally, we contribute to the literature on firms' responses to maternity-related policies. Much of this literature studies maternity benefits (Thomas, 2020; Fernández-Kranz and Rodríguez-Planas, 2021; Ginja et al., 2023; Ghosh et al., 2025); in the context of the PDA, Gruber (1994) studies the extension of health-insurance coverage to maternity and Timpe (2024) studies paid maternity-leave provisions. We document a trade-off between protecting the employment of pregnant women and the employment of women who *may become* pregnant, including those who will not. We do so in a particularly informative setting: the PDA-induced employment protection increased firms' exposure to underlying pregnancy and childbirth costs, in a context with scarce maternity benefits. Introductions and expansions of maternity benefits, on top of employment protection, may further increase fertility costs to firms, reinforcing the same trade-off depending on how benefits are designed and financed. Applying our framework to benefit expansions would provide complementary evidence on how specific changes in employer-borne fertility costs affect expected-fertility discrimination and its contribution to gender gaps.[1]

2 Conceptual Framework

We develop a framework to characterize expected-fertility discrimination and the challenges to its identification. We show how variation in employers' costs of future fertility can identify the discrimination it induces and compare this estimand with that obtained from variation in signals of future fertility. We then embed the framework in a stylized labor market model to characterize equilibrium employment and wage responses and their implications for the

[1]Quantification using wages is more difficult when workers highly value benefits, because wage declines may reflect both reduced labor demand and expanded labor supply (Summers, 1989). These forces affect employment in opposite directions, so an observed employment decline provides a lower bound on the magnitude of the employer response.

gender employment gap.

2.1 Expected-Fertility Discrimination and Identification

Our first goal is to establish whether employers differentially treat workers based on their expected future fertility. Consider a setting where workers differ by gender $g \in \{f, m\}$ and a vector of characteristics z, observed by employers. For each type (g, z), let $n(g, z)$ denote active headcount employment, $w(g, z)$ the wage, and $q(g, z) \in [0, 1]$ the employer's assessed probability of future pregnancy. Let $\mathbf{n} = n(g, z)_{g,z}$ collect employment across worker types.

We assume that output is given by a differentiable, increasing, and concave production function $Y(\mathbf{n})$ and output price is normalized to one. Costs come from wages and, conditional on future pregnancy, the present-value of all pregnancy and childbirth-related costs, $C \geq 0$. These costs may include replacement, accommodation, and benefit costs, productivity disruptions, or even non-pecuniary disutility associated with pregnancy, which could arise from biased preferences or paternalism.[2]

Taking wages and fertility expectations as given, the firm chooses employment to maximize

$$Y(\mathbf{n}) - \sum_{g,z} [w(g, z) + q(g, z)C]n(g, z). \tag{1}$$

The term $q(g, z)C$ is the expected-fertility cost associated with employing a worker of type (g, z). Because this term depends on the employer's expectation of future pregnancy, it can affect employment decisions before pregnancy occurs and can apply equally to workers who ultimately do and do not become pregnant but share the same observable characteristics. This is the source of expected-fertility discrimination.

We focus on hiring decisions as an illustrative example. Consider a firm whose current workforce is $\tilde{\mathbf{n}}$, which does not need to be optimal, and is deciding on a prospective hire with characteristics (g, z). The surplus from this marginal hire is

$$S(g, z; \tilde{\mathbf{n}}, C) \equiv MP(g, z; \tilde{\mathbf{n}}) - w(g, z) - q(g, z)C.$$

where $MP(g, z; \tilde{\mathbf{n}}) \equiv \partial Y(\tilde{\mathbf{n}})/\partial n(g, z)$ is the worker's marginal product. Identifying the presence of expected-fertility discrimination thus requires proving that hiring decisions are based on fertility expectations $q(g, z)$.

The main identification challenge is that employer's fertility expectations cannot be varied

[2]These costs account for the employer's optimal response once pregnancy occur. We discuss this in Section 3.1 in the context the choice of dismissing the worker upon pregnancy, which is directly influenced by the PDA.

independently of the characteristics from which they infer them. Consider exogenously varying an observable signal from z to z' for a prospective hire, holding gender, the firm's workforce, and fertility costs fixed. Even if the signal varies fertility expectations, $q(g,z) \neq (g,z')$, marginal productivity or wage may also vary between z and z'. Therefore, a resulting hiring response does not necessarily establish that the employer acts on fertility expectations.

Identification via variation in signals of future fertility requires additional assumptions: the signal z needs to be a valid instrument for fertility expectations.[3] It must be relevant for fertility expectations, so that $\Delta_z q \neq 0$, exogenously varied, and it needs to affect the employer's payoff only through those expectations, so that both $\Delta_z MP$ and $\Delta_z w$ are 0. Under these conditions, differences in hiring responses establish that employers discriminate based on expected fertility.[4] In practice, however, there is often a trade-off between relevance and exclusion. Characteristics that strongly predict fertility may also affect employers' beliefs about productivity or wage costs. Age, for instance, is highly correlated with fertility but also with work experience, and thus productivity. Indeed, for the set of worker characteristics that employers can generally observe, which are precisely the characteristics through which expected-fertility discrimination operates in reality, exclusion would not be a plausible assumption.[5]

Variation in the costs of fertility to the employer identifies expected-fertility discrimination, as operated by employers through the characteristics they actually observe. Consider an exogenous increase of the fertility costs from $C_0 \geq 0$ to $C_1 > C_0$. The hiring surplus will change by $-q(g,z)(C_1 - C_0)$. Hence, the increase reduces the return from hiring workers with positive expected fertility, proportionally to their assessed pregnancy probability, while leaving the surplus of workers with $q(g,z) = 0$ unchanged. In this controlled setting, where the increase in C can only affect this hiring decision, a differential hiring response by age or any other signal of future fertility, even if not excludable, would identify the presence of expected-fertility discrimination. The other difference with respect to variation in signals is that the cost shock identifies the *increase* in discrimination it induces. It can thus establish the presence of discrimination in the post-shock environment, but not generally the pre-existing presence of discrimination. Because of this, the external validity of the estimand depends on

[3]This view of z as an instrument has been discussed Rose (2023), where the inferred attribute is race, interpreted as a construct.

[4]No change in the hiring decision does not prove the absence of discrimination. This is because discrimination can be present and thus hiring surplus, S, change without passing the threshold for which the binary hiring decision changes.

[5]He et al. (2023) provide an example of a valid instrument, based on a characteristic usually not observed by employers. In their setting, whether applicants have no siblings becomes a predictor of future fertility due to a change in China's One-Child Policy. They show that providing this information to employers in a correspondence study reduces call-back rates for fertile-age women.

whether the fertility costs affected by the shock are relevant for other settings of interest.

In practice, variation in fertility costs is more likely to arise from policy changes than from experiments. Rather than individual hiring decisions in a controlled environment, we observe equilibrium responses of employment and wages. Since we are ultimately interested in the presence and consequences of discrimination in equilibrium, these responses are part of the object of interest.

Other applications: Although the equilibrium model developed below is tailored to expected-fertility discrimination, the identification challenge and the cost-based approach extend to other settings. For example, landlords may use tenants' observable characteristics to infer risks of nonpayment or property damage (Ewens et al., 2014; Arefeva et al., 2024), lenders may infer borrowers' default risk (Dobbie et al., 2021; Brock and De Haas, 2023), and employers may infer the risk of employee misconduct (Agan and Starr, 2018; Doleac and Hansen, 2020; Cullen et al., 2023). Such behaviors would constitute discrimination whenever protected characteristics, such as gender or race, are used to form expectations. Exogenous changes in landlords' exposure to tenancy losses, lenders' losses conditional on default, or employers' liability for misconduct could provide identifying variation in these settings. Although variation in individual characteristics has provided evidence of discrimination in some of these settings, variation in costs could offer important complementary evidence on the presence and consequences of these forms of discrimination in equilibrium. We next extend the framework to characterize these equilibrium responses in the context of expected-fertility discrimination.

2.2 Equilibrium Analysis

We embed the simple discrimination framework considered so far into a competitive model of the labor market with differentiated labor inputs, following Card and Lemieux (2001). In this section, we provide the main ingredients and results from the model, formal derivations are provided in Appendix A.2.

While employers may infer fertility from multiple characteristics, for the equilibrium analysis we focus on gender g and age z, distinguishing fertile-age workers ($z = 0$) and workers above childbearing age ($z = 1$). We assume that employers assign a positive probability of future pregnancy to fertile-age women, $q(f, 0) > 0$, and zero to older women and men, for whom $q(g, z) = 0$. We denote the inverse labor supply function of each type (g, z) as $W(g, z; n)$, which we assume to be a continuously differentiable and strictly increasing function of employment n.

We assume a nested production structure. Workers of different ages are imperfect substitutes within gender and are aggregated into female and male effective labor using CES functions:

$$L(g) = \left[\sum_{z=0}^{1}(a(g,z)n(g,z))^{\eta}\right]^{1/\eta}, \qquad a(g,z) > 0, \quad \eta < 1, \quad \eta \neq 0, \tag{2}$$

where $a(g,z)$ denotes the efficiency units supplied by each active worker of type (g,z), and $1/(1-\eta)$ is the elasticity of substitution across age groups. Output is produced according to $Y(\mathbf{n}) = \tilde{Y}(L(f), L(m))$, where $\tilde{Y}$ is increasing and concave. We leave the sign of the cross-partial $Y_{fm} \equiv \frac{\partial^2 \tilde{Y}}{\partial L(f)\partial L(m)}$ unrestricted, so female and male effective labor may be Edgeworth complements or substitutes. We analyze production through a single representative firm who chooses active labor inputs at market wages to maximize the objective in equation (1).

In an interior competitive equilibrium, firm's first-order conditions and market-clearing wages require that

$$MP(g,z;\mathbf{n}) = W(g,z,n(g,z)) + q(g,z)C, \qquad \forall\ (g,z). \tag{3}$$

The presence of expected-fertility discrimination means that, for fertile-age women, expected-fertility costs create a wedge between wages and marginal productivity. An increase in fertility costs raises this wedge, thereby increasing expected-fertility discrimination.

Equilibrium response to the cost shock. Consider a marginal increase in C, holding fertility expectations, technology, and labor-supply schedules fixed, while allowing employment and wages of all worker types to adjust. To characterize the equilibrium response, define

$$\mathcal{J}(g,z) \equiv \frac{\partial W(g,z;n(g,z))}{\partial n(g,z)} - \frac{\partial MP(g,z;\mathbf{n})}{\partial n(g,z)} > 0,$$

which captures the slopes of labor supply and marginal productivity with respect to the type's own employment and is positive given the assumptions. Differentiating the equilibrium conditions given by (3) we obtain

$$\frac{dn(g,z)}{dC} = \frac{-q(g,z) + \sum_{k\neq(g,z)} \frac{\partial MP(g,z;\mathbf{n})}{\partial n_k}\frac{dn_k}{dC}}{\mathcal{J}(g,z)}. \tag{4}$$

where k indexes types other than (g,z). The first term in the numerator captures the direct fertility-cost shock, while the second captures changes in marginal productivity induced by

employment adjustments among other worker types. For workers of the opposite gender, the sign of this marginal-product interaction is determined by Y_{fm}. Thus, expected fertility determines direct exposure to the shock, but not necessarily the magnitude or sign of the equilibrium employment response. Since labor supply is upward sloping, wages and employment move in the same direction within each cell.

For fertile-age women, the equilibrium response is unambiguously negative. In equilibrium, adjustments in other labor inputs can attenuate or reinforce the direct fertility-cost shock. Nevertheless, as shown in Appendix A.2, an increase in C necessarily reduces pregnancy-risk-weighted employment. That is, the amount of employment weighted by each group's probability of pregnancy. A standard law-of-demand logic applies: because the increase in C raises expected labor costs in proportion to pregnancy risk, pregnancy-risk-weighted employment must decline in equilibrium. Since in this stylized framework $q(g, z)$ is positive only for fertile-age women, their employment must decline as C increases. Hence,

$$\frac{dn(f,0)}{dC} < 0, \qquad \frac{dw(f,0)}{dC} < 0.$$

As equation (4) shows, workers with no direct fertility-cost exposure can also be affected through equilibrium spillovers. Older women's employment and wages may rise or fall. The contraction among fertile-age women reduces older women's marginal product within the female CES aggregate $L(f)$, but also makes female effective labor scarcer, potentially raising its marginal value Y_f. Thus, although female effective labor necessarily declines, $dL(f)/dC < 0$, total female headcount employment remains ambiguous. Male responses depend on the technological interaction between female and male effective labor. Male employment and wages fall when female and male effective labor are Edgeworth complements ($Y_{fm} > 0$), rise when substitutes ($Y_{fm} < 0$), and remain unchanged when $Y_{fm} = 0$.

Contribution to the gender employment gap. Define the gender employment gap among fertile-age individuals at given fertility costs C as $\Gamma(C) \equiv n(m, 0; C) - n(f, 0; C)$, where $n(g, 0; C)$ denotes equilibrium employment given C.[6]

An increase in fertility costs can both establish the presence of expected-fertility discrimination after the cost shock and quantify the contribution of the additional discrimination it induces to the gender employment gap. An increase from C_0 to C_1 changes the gap by $\Gamma(C_1) - \Gamma(C_0)$. To measure its relative contribution to the gap observed *after* the shock, which

[6]Although in the empirical analyses we focus on the gap in employment-to-population ratios, here we abstract from population sizes, which are not needed for the equilibrium analysis. The results for gaps in ratios are equivalent to those in this section, under the assumption that fertile-age populations are equal across genders and unaffected by C.

includes the induced increase in discrimination, we divide this difference by the post-shock gap:[7]

$$\mathcal{S}(C_1, C_0) \equiv \frac{\Gamma(C_1) - \Gamma(C_0)}{\Gamma(C_1)}. \tag{5}$$

Although fertile-age women's employment necessarily declines, the gender gap may widen or narrow because male employment may also fall. That is, *ex ante*, an increase in expected-fertility discrimination does not necessarily widen the gender gap. Its response must therefore be established empirically, which we do in Section 7, documenting that the contribution is indeed positive and large.

Finally, the contribution identified by the cost shock, $\mathcal{S}(C_1, C_0)$, under certain conditions, provides a lower bound for the overall contribution of expected-fertility discrimination. Consider the counterfactual with no fertility costs, where the expected-fertility discrimination is thus absent. The overall contribution of this form of discrimination can be expressed as the $\mathcal{S}(C_1, 0)$, the share of the gender gap observed after an increase in costs from 0 to C_1 that is due to the extent of discrimination induced by this increase. Assuming that employment is higher for men than for women at C_1, it follows that

$$\mathcal{S}(C_1, 0) \geq \mathcal{S}(C_1, C_0) \quad \text{if and only if} \quad \Gamma(C_0) \geq \Gamma(0). \tag{6}$$

This condition is not guaranteed because pre-existing discrimination may also reduce male employment relative to the counterfactual with no discrimination, potentially narrowing the gender gap. While the effects identified by the cost shock cannot establish whether this condition holds, we combine them with parameters from the literature to calibrate the model and evaluate its validity empirically in Section 7.

3 Institutional Background

In this section, we provide a brief overview of the legal history of pregnancy discrimination at the state and federal level up to the passage of the Pregnancy Discrimination Act in 1978. We conclude by discussing how the Act provides a source of plausibly exogenous variation in the costs of fertility to the firm, allowing us to identify expected-fertility discrimination as described in Section 2.

Already in the 1960s, legislators had addressed gender discrimination in the labor market with the Equal Pay Act (EPA) of 1963 and Title VII of the Civil Rights Act of 1964. The former forbade sex-based wage discrimination, while the latter extended this protection to all

[7] Dividing by the baseline gap $\Gamma(C_0)$ would measure the relative magnitude of the *effect* on the gap, not its contribution.

aspects of employment and also to discrimination based on race, color, religion, and national origin. However, neither of these acts explicitly recognized pregnancy discrimination as a form of gender discrimination, leaving a gap in the federal equal-treatment mandate.

In 1973, the EEOC tried to address this gap by promulgating guidelines specifying that discrimination on the basis of pregnancy constituted gender discrimination[8]. While some lower federal courts initially accepted this view, it was not the prevailing interpretation of federal law (Wolkinson and Liberson, 1982). This narrower view was upheld in two Supreme Court decisions concerning disability benefits for pregnancy. In *Geduldig v. Aiello* (1974) and *General Electric v. Gilbert* (1976), the Court ruled that pregnancy-based distinctions were not, by themselves, gender-based classifications (Pedriana, 2009)[9]. Thus, before 1978, pregnancy discrimination was not recognized as gender discrimination under federal law.

At the state level, however, pregnancy protections had already followed a different path. Before 1978, 19 states and the District of Columbia had adopted policies concerning the treatment of pregnant workers. These policies varied both in their provisions and mode of passage—through acts of legislation, administrative rulings, and state supreme court decisions. Ten jurisdictions formally mandated equal treatment of pregnancy and temporary disabilities, as the PDA would subsequently require[10]. Six states strengthened employment protection further by requiring employers to provide job-protected unpaid leaves of reasonable length[11]. Importantly, many of these policies predated the Supreme Court rulings of 1974 and 1976, and they continued to be enforced and regarded as valid in the respective states after those decisions (Wolkinson and Liberson, 1982). Finally, five states addressed the unequal treatment of pregnancy in their short-term temporary disability insurance (STDI) programs by including pregnancy among qualifying conditions[12]. These reforms required pregnancy-related disabilities to be treated like other temporary disabilities for the purpose of income replacement, but did not mandate equal treatment in other dimensions of employment

[8]U.S. EEOC, "Guidelines on Discrimination Based on Sex," 29 CFR 1604.10 (1973).

[9]The main rationale was that groups divided by gender and by pregnancy status did not entirely overlap, as both men and women can be in the *non-pregnant* group. Footnote 20 of the majority opinion in *Geduldig v. Aiello* (1974) reads: 'The program divides potential recipients into two groups – pregnant women and non-pregnant persons. While the first group is exclusively female, the second includes members of both sexes. The fiscal and actuarial benefits of the program thus accrue to members of both sexes" (see here). In *Cleveland Board of Education v. LaFleur* (1974), the Supreme Court ruled against public-school policies that forced pregnant teachers to leave their positions for five months around childbirth. However, this was based on the Due Process Clause, rather than on equal protection grounds. As Pedriana (2009) states, "*LaFleur* did not directly address whether pregnancy discrimination was tantamount to sex discrimination."

[10]Pennsylvania in 1973; New York in 1974; Alaska, Iowa, South Dakota, and Wisconsin in 1975; and Maryland, Minnesota, Oregon, and the District of Columbia in 1977.

[11]Colorado, Kansas, Massachusetts, and Washington in 1972; Connecticut in 1973; and Montana in 1975.

[12]Rhode Island in 1942; New Jersey in 1961; Hawaii in 1973; California in 1976; and New York in 1977. New York therefore belongs to both groups.

and, in particular, did not provide employment protection. Thus, before 1978, 16 jurisdictions had adopted employment protections comparable to or stronger than the PDA, while four additional states had addressed pregnancy only through STDI programs.

The emergence of these state policies was part of the broader expansion of women's rights and state civil-rights institutions during this period. One manifestation of states' receptiveness to this agenda was their support for the federal Equal Rights Amendment (ERA), which sought to establish constitutional equality of rights irrespective of sex and, at the time, enjoyed broad bipartisan support. Support for the ERA was particularly widespread among the states that adopted pregnancy protections: all 15 formal adopters had ratified it by 1977, and 14 had done so before adopting their pregnancy policy.[13] Yet ERA ratification was far from unique to these states, as many states without comparable pre-1978 pregnancy protections also ratified the amendment. Early adoption therefore did not simply reflect a uniquely favorable attitude toward women's equality. Rather, the historical evidence points to the interaction of broader political receptiveness with other institutional and political conditions—including women's-rights mobilization, supportive political elites, and access to legislatures, civil-rights agencies, or state courts—that allowed these claims to be translated into binding protections (Ramanathan, 2021). In some states, the Supreme Court decisions further increased the pressure to formalize these protections, contributing to the adoption or codification of rules that rejected the narrower federal interpretation.

The Supreme Court decisions also intensified demands to address this gap at the federal level. A coalition of civil-rights organizations, women's-rights groups, and labor unions viewed the rulings as fundamentally undermining the protections against sex discrimination mandated by Title VII. As Susan Ross, on behalf of the Coalition to End Discrimination Against Pregnant Workers, stated in 1977: "Employers routinely fire pregnant workers, refuse to hire them, strip them of seniority rights, and deny them sick leave and medical benefits given other workers. [... E]mployers use women's role of childbearer as the central justification of and support for discrimination against women workers. Thus, discrimination against women workers cannot be eradicated unless the root discrimination, based on pregnancy and childbirth, is also eliminated"[14]. The issue was becoming increasingly salient as employment among women, particularly fertile-age women, continued to rise, while men's employment remained high, but comparatively flat (Blau, 2025). This context pushed Congress, which at the time had large Democratic majorities, and the Carter administration to address the issue legislatively at the federal level, passing the PDA on October 31, 1978.

[13]Washington is the exception: it adopted its pregnancy regulation in 1972 and ratified the ERA in 1973.

[14]Hearings on S. 995 before the Subcommittee on Labor of the Senate Committee on Human Resources, 95th Cong., 1st Sess. (1977).

The PDA mandated that women affected by pregnancy, childbirth, or related medical conditions be treated equally to other workers similar in their ability or inability to work, applying to establishments with 15 or more employees.[15] *De jure*, this forbade employment discrimination on the basis of pregnancy in all aspects of employment. *De facto*, other than its extension of employer-provided health-insurance benefits in a specific set of cases[16], the PDA effectively established a form of employment protection for pregnant workers at the federal level. As in analogous contexts[17], discrimination at the hiring stage was prohibited but substantially harder to detect and prove in court than dismissal after pregnancy became observable. This interpretation is consistent with patterns in EEOC charge statistics. First, among sex-discrimination charges against private employers, dismissal was already a relatively common issue before the PDA, but its share rose sharply in 1978 and it became the single most frequently reported issue thereafter. From 1978 onward, dismissal accounted for roughly one-third of all sex-basis issue allegations, substantially more than any other separately reported issue (Figure A1). Second, among pregnancy-discrimination cases pursued after the PDA, only those concerning discharge exhibited a consistently high rate of decisions favoring workers (Edwards, 1996). Anecdotally, notable legal cases following the PDA also show employers attempting to "hide" dismissals upon pregnancy, suggesting awareness of the sanctions imposed on unjust dismissal.[18]

Finally, the Act did not introduce special protections for pregnant workers but instead mandated *equal* treatment relative to temporary disabilities.[19] Hence, it protected the employment of pregnant women only in firms that protected the employment of temporarily disabled workers, often through job-protected unpaid leaves. The effect of the PDA that we identify is therefore the effect of this *equal-treatment mandate*, rather than employment protection for *all* pregnant workers. For brevity, we refer to the effects of the PDA as the effects of mandated employment protection of pregnant women, but this distinction should be kept in mind. It also distinguishes the PDA and comparable state policies from pregnancy coverage under state STDI programs, which provided income replacement without requiring employers to protect the worker's job.[20]

[15]Full text available here, and the associated EEOC guidelines here.

[16]This aspect of the Act has been studied by Gruber (1994) and applied only to establishments with certain insurance plans in place.

[17]E.g., disability discrimination and the Americans with Disabilities Act (Acemoglu and Angrist, 2001).

[18]See *Tamini v. Howard Johnson* (1987) and *EEOC v. Service News Co.* (1991).

[19]A large body of legal literature has studied an unintended consequence of the PDA: mandating equal treatment could hinder employers' and states' attempts to provide special protections and benefits to pregnant workers that were not granted to other temporarily disabled workers (Remmers, 1989; *California Federal S. & L. Assn. v. Guerra*, 479 U.S. 272 (1987)).

[20]Timpe (2024) studies the extension of temporary disability benefits to pregnancy under state STDI programs and employer-provided disability plans, which provided income replacement during absences without

The complete timing, legal basis, and provisions of the pre-1978 state policies are reported in Appendix Table A2 and summarized by Figures A3 and A2. Appendix A.1 provides more details on the sources used to compile and classify this list of policies. Section 5 discusses how we use this variation in the empirical analysis.

3.1 The PDA as a Shock to Fertility Costs

Having motivated why we mainly interpret the PDA as increasing employment protection for pregnant workers, we now map this change into the fertility costs C introduced in Section 2.

When a worker becomes pregnant, the employer can either dismiss her, at an effective present-value cost ϕ, inclusive of expected legal costs, or retain her, at a present-value cost R. This cost may include temporary replacement costs, costs of workplace accommodation, and maternity benefits, although generally rare in this setting. We assume that workers do not contribute to active production while pregnant. The optimized cost of future fertility is therefore $C(\phi) = \min\{\phi, R\}$. Raising employment protection, the PDA increases ϕ, say from ϕ_0 to ϕ_1, and thus fertility costs for firms that would choose to dismiss pregnant workers. In our stylized framework, where firms are homogeneous, we assume all them to be dismissing pregnant workers at the initial level ϕ_0.[21] The magnitude of the increase in C depends on whether dismissal remains optimal after the PDA. If so, the PDA increases costs by $\Delta C = \phi_1 - \phi_0$. If firms switch to retention, $\Delta C = R - \phi_0$. Importantly, fertility costs increase in both cases. As a tractable baseline, motivated by the historical evidence of continued dismissals after the PDA, we maintain the first case in the framework.[22] Since fertility costs may already have been positive before the PDA, ΔC identifies an increase in expected-fertility discrimination.

The PDA provides a plausibly valid and exogenous source of variation in fertility costs, relevant beyond this particular policy. First, to be a valid cost shock, as the ones considered in Section 2, the policy should affect labor market outcomes only through its effect on firm-borne fertility costs. By design, the PDA primarily changed how employers could respond to a worker's pregnancy, which supports this assumption. In Appendix A.2.4, we

guaranteeing job protection. We are very grateful to Brenden Timpe for sharing his notes on these policies.

[21] In practice, firms already retaining pregnant workers and firms dismissing all temporarily disabled workers should not be affected by the increase in ϕ. The estimated response to the PDA will thus be average of directly affected firms that before were discriminatorily dismissing and of those that were not. Since we are interested in studying discrimination in equilibrium, these are the relevant estimates.

[22] Under retention, the main difference is that the model needs to distinguish active employment from retained pregnant workers. For the empirical analyses, the increase in retention would raise the number of pregnant women counted as employed, partly offsetting the overall decline in fertile-age women's employment. We thus capture the net effects of lower employment due to increased expected-fertility discrimination and higher employment due to increased retention upon pregnancy, amounting to an underestimate of the effects of discrimination. Appendix A.2.3 extends the model to formally account for this.

discuss potential violations, including changes in fertility expectations, labor supply, and maternity benefits. We argue that these alternative channels either also operate through expected-fertility discrimination, even if not only through fertility costs, or tend to attenuate its negative employment effects. Second, the PDA generated differential exposure to increased fertility costs across states with and without comparable pre-existing protections. We discuss why this variation is plausibly exogenous to counterfactual labor market trends in Sections 5 and 6. Third, the increase in expected-fertility discrimination identified through the PDA is informative about a broad range of settings, particularly today, when maternity-related protections are more common. Maternity benefits were rare at the time and generally not expanded by the PDA. The discriminatory response therefore primarily reflects the increased cost of avoiding expenses associated with retaining pregnant workers, including replacement, accommodation, and productivity disruptions. Since firms could always switch to retention, the PDA could at most make them bear these costs in full. While the exact magnitude of the response may differ, the same source of discrimination is therefore relevant in all settings where firms bear these costs rather than avoid them through dismissal.

4 Data

Our main analyses combine two individual-level surveys, the Current Population Survey Annual Social and Economic Supplement (CPS ASEC) and the Panel Study of Income Dynamics (PSID), with several complementary data sources.

CPS ASEC. The CPS ASEC is a nationally representative annual supplement to the March CPS. Its main advantage for our purposes is that it collects detailed information on labor-force status and earnings over the previous calendar year. Our main outcome is an indicator for whether an individual was employed at any point during that year. We prefer this measure to employment in the survey week because it is less sensitive to short-term fluctuations, but show robustness to the latter. Importantly, our outcome distinguishes employment from non-employment rather than employment from unemployment, and therefore also captures movements into and out of the labor force, an important margin for women during this period. Until 1976, the CPS ASEC groups some states of residence together in clusters; we discuss the resulting implications for our empirical design in Section 5.

We construct proxies for hiring and firing from additional employment questions. We define hiring as reporting employment in the survey week but no employment during the preceding year. Because the ASEC is conducted in March, this measure captures only a subset of hiring events, and we therefore prioritize the more precise PSID measure described

below. We proxy firing using reported layoffs among respondents absent from work in the survey week. We construct hourly wages as annual wage and salary income divided by annual hours worked, where annual hours are the product of weeks worked and usual weekly hours. This measure is available since 1975.

Finally, we construct a proxy for pregnancy using information on the presence and age of children. A woman is classified as having been pregnant during the previous year if she reports a child less than one year old at the time of the March survey. We additionally use standard demographic and employment characteristics, including age, race, education, marital status, location, and industry of employment. Throughout, variables referring to the previous calendar year are assigned to that reference year rather than to the year in which the survey was conducted.

PSID. We complement the CPS ASEC with the PSID (Pfeffer et al., 2025; Panel Study of Income Dynamics, 2026), a longitudinal household survey following families and their descendants since 1968. Most variables are defined analogously to those in the CPS. We provide details in Appendix A.4.

Because we observe individuals across years, we can measure hiring from changes in annual employment status rather than from employment during a single survey week. We can also measure job separations and their reasons, although for women this information is mostly available since 1979. The PSID additionally provides information on completed fertility and respondents' childhood state of residence, which we use in robustness and heterogeneity analyses. Our analyses exploit the repeated cross-sectional variation in the PSID and include all respondents, including so-called "nonsample persons".[23]

Additional data. We use four additional sources for complementary analyses. First, the Basic Monthly CPS provides higher-frequency measures of employment and employment transitions, which we use to examine dynamics around the enactment of the PDA and as a robustness check. Second, we use Census microdata. The 1% Form 1 state sample of the 1970 Census provides a more precise measure of female employment shares across industries, while the 1% and 5% samples of the 1960 and 1980 Censuses, respectively, are used in additional robustness exercises. Third, we use the 1976 Survey of Income and Education (SIE) (US Bureau of the Census, 1992), a large CPS supplement, to construct industry-level measures of the availability of health and temporary-disability insurance. Finally, we

[23]Nonsample persons are respondents who were not part of the original 1968 sample but subsequently entered households belonging to original sample families. Because the PSID provides survey weights for these respondents only beginning in 1997, we construct weights for earlier years by extending the methodology in Berglund and Heeringa (2015).

use County Business Patterns (CBP) as an administrative source of information on local economic activity. Unlike our survey data, CBP does not distinguish employment by workers' demographic characteristics. We use CBP data and state GDP estimates from the Bureau of Economic Analysis (BEA) to document aggregate effects on employment, establishments and output.

5 Empirical Strategy

Section 2 showed how the effects of the PDA can be used to establish the presence of expected-fertility discrimination. We now discuss how we identify and estimate the causal effects of the PDA.

Research design. Because the PDA was a federal law, it applied nationwide. We therefore construct a comparison group using states that had already enacted policies providing protections comparable to those subsequently introduced by the PDA, as discussed in Section 3. We refer to states first subject to these protections through the PDA in 1978 as *PDA-treated* states, and to states with comparable pre-existing protections as *PDA-control* states. The identifying variation comes from comparing the change in outcomes around 1978 in PDA-treated states with the corresponding change in PDA-control states, whose treatment status plausibly did not change with the passage of the PDA.[24]

Baseline specification. We implement this comparison using the following difference-in-differences specification:

$$y_{i,s,t} = \beta T_s \times P_t + \Gamma X_{i,s,t} + \theta_s + \delta_t + \varepsilon_{i,s,t}, \tag{7}$$

where $y_{i,s,t}$ is the outcome of interest for individual i residing in state s in year t. T_s equals one for PDA-treated states and zero for PDA-control states, while P_t equals one from 1979 onward and zero otherwise. We begin the post-treatment period in 1979 because the PDA was enacted on October 31, 1978, making 1979 the first full year of treatment.[25] $X_{i,s,t}$ includes a polynomial in age, marital status, race, education, and metropolitan-area residence. θ_s

[24] An alternative would be to exploit the earlier staggered adoption of state pregnancy protections directly. We present descriptive evidence from these reforms in Appendix Figure A6, which is consistent with what we document using the PDA, but do not use them as our preferred design. State-level samples are small, and some early policies provided stronger protections than the PDA, including job-protected unpaid leave, making their effects imperfect analogues of the federal reform.

[25] This is a conservative choice: any 1978 effect would mechanically attenuate our estimates.

and δ_t denote state and year fixed effects, respectively.[26] We report heteroskedasticity-robust standard errors in the baseline. Because treatment varies at the state level and the number of clusters is small, Appendix A.7 assesses robustness using conventional clustered inference and alternative procedures designed for settings with few clusters.

We estimate equation (7) for fertile-age women, our main group of interest, as well as for the other demographic groups. The coefficient β identifies the average treatment effect of the PDA on the treated, where the treated are individuals residing in PDA-treated states, under the identification assumptions described below.

The main identification assumption is that, absent the PDA, outcomes in PDA-treated and PDA-control states would have followed parallel trends around 1978. Because PDA-control states were already subject to pregnancy protections, this assumption also requires that the effects of their pre-PDA policies had stabilized by the period relevant for our comparison. In particular, the contribution of these earlier policies to outcomes in PDA-control states must no longer be evolving around 1978. If the effects of the earlier policies were still growing or decaying, the resulting dynamics would enter the outcome trend of PDA-control states and bias the comparison. We evaluate the plausibility of this assumption using both raw outcome trends and the event-study specification described below.

A second requirement is that the treatment status of individuals in PDA-control states did not change when the PDA was enacted. For this reason, we use as PDA-controls only states whose pre-existing laws provided protections comparable to, or stronger than, those introduced by the PDA, as documented in Section 3 and Appendix A.1. The federal law therefore did not introduce new protections for workers in these states. It remains possible that federal enforcement by the EEOC after 1978 increased the effective intensity of protection even in PDA-control states. To the extent that such an increase affected outcomes in the same direction as the PDA in PDA-treated states, it would bias our estimates toward zero.

Finally, as in a standard difference-in-differences design, identification requires the absence of spillovers from the PDA across PDA-treated and PDA-control states. We examine the sensitivity of the results to this concern in Section 6.4.

Event-study specification. We also estimate a fully dynamic version of equation (7):

$$y_{i,s,t} = \sum_{l \neq 1977} \beta_l T_s \times \mathbf{1}\{t = l\} + \Gamma X_{i,s,t} + \theta_s + \delta_t + \varepsilon_{i,s,t}, \tag{8}$$

[26] For the CPS ASEC, s corresponds to the geographic unit identified in the data, which can be either an individual state or a group of states. See Section 4 for details.

where 1977, the last full pre-PDA year, is the omitted reference period. The coefficients β_l measure the difference between PDA-treated and PDA-control states in year l relative to their difference in 1977.

The pre-PDA coefficients assess whether the two groups were following parallel trends before the reform. This is particularly informative in our setting because differential pre-trends could reflect either differences in underlying outcome trends or continued dynamics in the effects of the earlier policies in PDA-control states. The absence of differential pre-trends therefore provides evidence in favor of the identifying assumption that the two groups would have continued to evolve in parallel around 1978 absent the PDA.

The post-PDA coefficients characterize the dynamics of the treatment effect in PDA-treated states. The coefficient for 1978 captures the transition year, while coefficients from 1979 onward correspond to full-treatment years. A treatment effect that stabilizes relatively quickly after the PDA also provides evidence on the speed of adjustment to these policies and, in turn, on the plausibility that the effects of the earlier policies in PDA-control states had stabilized within the time window we consider.

Choice of states and time window. The construction of the baseline sample reflects both the identification requirements of the design and limitations of the CPS ASEC. We restrict the comparison group to states that adopted qualifying pregnancy protections no later than 1973. Focusing on sufficiently early adopters increases the time between the introduction of their policies and the PDA, making it more plausible that the effects of these policies had stabilized by 1978. Later-adopting states are less suitable controls because their treatment effects may still have been evolving around the federal reform. We therefore exclude them from our preferred specification. We use a 1973–1984 time window centered around the passage of the PDA.

As mentioned in Section 4, the CPS ASEC identifies some states only as part of multi-state groups during our sample period. We retain PDA-control states only when the corresponding CPS geographic unit can be assigned unambiguously to the control group, and similarly retain PDA-treated states only when they are separately identified or grouped exclusively with other states first treated by the PDA. This ensures that every observation can be assigned an unambiguous treatment status.

The resulting PDA-treated states in our preferred specification are Alabama, Arkansas, Florida, Georgia, Illinois, Indiana, Kentucky, Louisiana, Mississippi, North Carolina, Ohio, Oklahoma, South Carolina, Tennessee, and Texas. The PDA-control states are Connecticut, Massachusetts, and Pennsylvania. Figure 1 displays the resulting geographic comparison. Because these groups differ geographically and the number of suitable early-treated states is

necessarily limited, we check that results are not driven by this particular choice of states.

Alternative designs and robustness. Section 6.4 assesses the sensitivity of the baseline estimates to alternative definitions of the PDA-treated and PDA-control groups and to alternative identification strategies. In particular, because the PSID identifies states separately, it allows us to use a substantially larger set of states than the CPS ASEC; Figure A9 displays the resulting broader geographic comparison. We also consider specifications that exploit additional sources of variation, including triple-difference designs. Together, these exercises assess whether the baseline estimates are sensitive to the CPS-imposed geographic composition of the treatment and control groups or to differential trends across these groups for reasons unrelated to the PDA.

Main Sample and Summary Statistics Our main group of interest is fertile-age women, which we define as aged 18 to 35 based on the fertility patterns observed at the time (Fig. A4).[27] Following our framework, we also consider women aged 40 to 60 and men aged 18 to 60 as potentially indirectly affected groups. Table 1 reports summary statistics for these groups using the full 1973–1984 sample across PDA-treated and PDA-control states.

Among women aged 18–35, 81% have completed high school, 35% have attended college, 83% are white, and 61% are married. Older women have substantially lower educational attainment, while men have higher rates of college education. Women aged 18–35 have on average 1.1 children at the time of the survey, compared with 2.8 among older women and 1.6 among men. In the CPS ASEC, 72% of women aged 18–35 are employed, compared with 60% of older women and 91% of men; the corresponding PSID rates are 61%, 58%, and 86%. Average hourly wages for younger women are \$4.86 in the CPS ASEC and \$5.16 in the PSID, compared with \$7.83 and \$8.56 for men. Differences across the two datasets partly reflect differences in sample and variable construction; Appendix A.4 provides details.

About 8.5% of women aged 18–35 experience a pregnancy in a given year, and about 56% are employed during pregnancy, which is high considering that the same rate is 61–72% among fertile-age women overall. This also highlights how employment protections around childbirth were directly relevant for a substantial share of fertile-age women. Among those we can follow in the PSID, about 42% subsequently separate from their job. Most of these separations are recorded as "voluntary," although this category also includes pregnancy-related separations and may therefore capture part of the phenomenon the PDA was intended to address. Firings and layoffs alone account for about 9% of post-pregnancy separations:

[27] This cutoff captures the ages at which births are concentrated while maintaining a clear separation from our comparison group of older women.

3.8% of recently pregnant women report being fired or laid off, compared with only 0.8% of women and 0.7% of men aged 18–35 overall. Importantly, these separation statistics are based predominantly on the post-PDA period, when detailed PSID information on the timing and reasons for separation is available. Thus, even with the PDA already in place, employer-initiated separations following pregnancy were far from negligible, especially compared to their prevalence in the broader workforce.

6 Results

We now estimate the effects of the PDA empirically. We first establish the employment effects on fertile-age women and show how they align with the expected-fertility discrimination mechanism presented in Section 2. We then examine adjustments in hiring, wages, and retention of pregnant workers, as well as equilibrium spillovers to men and older women.

6.1 Employment Response and Expected-Fertility Discrimination

The raw employment trends show a relative decline in fertile-age women's employment in PDA-treated states after 1978. Figure 2 plots quarterly employment rates from the monthly CPS for women ages 18–35 in PDA-treated and PDA-control states. Employment is rising in both groups before the reform and the two series evolve similarly, with employment in PDA-treated states consistently at higher levels. This pattern changes exactly after the passage of the PDA: employment in PDA-treated states declines relative to PDA-control states and remains at a slightly lower relative level thereafter.

The longer-run trends show a relative employment decline in PDA-control states following their earlier adoption of pregnancy protections, with the pattern reversing after the PDA. Appendix Figure A6 extends the window backward using the CPS ASEC.[28] The two groups follow broadly parallel trends before the PDA-control states introduce their own pregnancy protections earlier in the 1970s. After these policies are enacted, employment in PDA-control states deviates from this common trend in the direction predicted by our framework. This relative pattern subsequently reverses when the PDA extends comparable protections to the remaining states in 1978.

The event-study estimates confirm a persistent decline in fertile-age women's employment in PDA-treated states relative to PDA-control states after 1978. Figure 3 reports estimates from specification (8) using the CPS ASEC and PSID. In the CPS ASEC, the coefficients

[28] Massachusetts cannot be included in this figure since it is not separately identified prior to 1973. As a result, trends in PDA-control states in this figure are slightly different and noisier.

before 1978 are small and provide no evidence of differential pre-trends between PDA-treated and PDA-control states. Beginning after the passage of the PDA, the coefficients become persistently negative and economically sizable, confirming the relative employment decline visible in the raw data. The corresponding PSID estimates display the same post-PDA decline, although the yearly estimates are considerably noisier given the smaller sample.

In both datasets, the effect emerges within a year of the PDA and remains relatively stable thereafter, suggesting that the employment response to the federal reform had little subsequent dynamic adjustment. To the extent that the earlier state pregnancy protections generated similarly rapid and persistent effects, the PDA-control states would have already reached a stable post-treatment path by 1978. This further supports their use as a comparison group. Thus, the event-study evidence both confirms a persistent decline in employment after the PDA and supports the identifying assumptions underlying our design.

The aggregate DID estimates indicate that the PDA reduced employment of fertile-age women by 4.5 percentage points in the CPS ASEC, a 6.6% decline relative to the pre-PDA employment rate of 68.6% (Table 2). The PSID confirms this result, with a decline of 6.9 percentage points (10.5% relative to its pre-PDA mean; Table 3). The monthly CPS yields a smaller but precisely estimated decline of 3.0 percentage points (Table A3).[29] The negative employment effect is robust to alternative inference procedures designed for settings with few clusters (Table A13).

The magnitude of the employment decline is large, but consistent with evidence from other policies that raise the cost of employing a particular subgroup. Our preferred estimate of a 6.6% relative decline is close to the 5.1% decline documented by Fernández-Kranz and Rodríguez-Planas (2021) in response to a policy that increased firms' expected costs associated with fertility by granting parents stronger rights to reduce working hours after childbirth. Similar magnitudes are found in other settings where increased labor-costs disproportionately affect particular demographic groups: Marchingiglio and Poyker (2026) estimate a 3.1% decline in female employment following gender-specific minimum-wage legislation, Bailey, DiNardo, et al. (2021) find a 3.4% decline in employment among Black men, for which the minimum-wage floor was more binding. Two features of our setting help rationalize the larger magnitude we document: female employment elasticity was high at the time (Elder et al., 2025), and the PDA introduced a new federal protection rather than marginally expanding an existing mandate, potentially implying a larger shock to employers' expected costs.

The employment decline is concentrated at ages with higher fertility risk, consistent with the expected-fertility discrimination mechanism predicted by our framework. In Figure 4, we

[29]The monthly CPS measures current employment, the CPS ASEC and PSID measure whether the respondent was employed at any point during the year. Magnitudes are thus not directly comparable.

estimate the effect of the PDA separately across narrow age groups and plot these estimates alongside the fertility rate at each age, also observed from the CPS ASEC. The employment decline is concentrated at ages at which pregnancy risk is economically relevant. As fertility risk becomes small, the estimated employment effect converges toward zero and eventually disappears.

This pattern corresponds closely to the exposure mechanism in the model. Pregnancy risk $q_f(z)$ determines exposure to the direct cost shock: workers with $q_f(z) = 0$ are not directly affected, whereas workers with $q_f(z) > 0$ become more costly to employ. The model does not require the equilibrium employment response to be proportional to $q_f(z)$, because age-specific labor-supply responses, productivity, and interactions across labor inputs also determine the magnitude of the equilibrium adjustment. However, the model predicts that the negative response should be concentrated where fertility risk is positive, which is what we observe. Importantly, the analogous age profile for men does not follow their fertility profile (Appendix Figure 4b).

We obtain similar evidence when we use a broader set of individual characteristics to predict future fertility. Using the PSID, we estimate the probability of having a child in the following one to two years as a function of pre-determined worker characteristics and apply the estimated prediction model to classify women in the CPS ASEC as having high versus low predicted fertility based on their characteristics.[30] Appendix Figure A7 shows that the employment decline is concentrated among women with high predicted fertility, whereas there is little corresponding difference by predicted fertility among men.

6.2 Margins of Adjustment

We next examine outcomes beyond employment to understand the margins through which this response occurred.

Hiring. The PDA reduced hiring of fertile-age women in both the PSID and monthly CPS, although the estimates are imprecise. Measuring hiring requires observing employment transitions over time, so we rely primarily on these two datasets, which follow respondents across survey waves. In the PSID, the probability of moving into employment declines by 5.3 percentage points, or 19% relative to the pre-PDA mean (Table 3). The monthly CPS yields a decline of 0.5 percentage points, relative to a pre-PDA monthly hiring rate of 6% (Table A3). For completeness, Table 2 also includes a hiring proxy from the CPS ASEC, which yields an estimate close to zero. As discussed in Section 4, this measure captures only

[30] We describe this measure and exercise in detail in Appendix A.6

a narrow subset of hires occurring between January and March, making it less informative about the overall hiring response. Taken together, the evidence suggests that reduced hiring was an important margin of adjustment in fertile-age women's employment.

Wages. Adjustment through wages is considerably smaller. In the CPS ASEC, the estimated effect on hourly wages is -0.021 standard deviations and is not statistically significant. In the PSID, the coefficient is close to zero. Unlike the employment estimates, however, these coefficients do not have a straightforward causal interpretation: wages are observed only among employed workers, and the PDA itself changes selection into employment. Nonetheless, the fact that average observed wages change little in the face of a sizeable employment decline indicates that adjustment on the wage margin was relatively limited compared with employment.

Two forces may help explain this pattern. First, labor supply among fertile-age women was relatively elastic at the time (Elder et al., 2025). In line with our framework, this makes a decline in labor demand translate disproportionately into lower employment rather than lower equilibrium wages. Second, wage rigidities may have further limited firms' ability to adjust on the wage margin. Relative to the flexible-wage benchmark in our framework, constraints on downward wage adjustment shift more of the response to a labor-demand shock onto employment and can therefore exacerbate the employment decline generated by the increase in fertility costs.

One potentially important source of such rigidities in this setting is the Equal Pay Act of 1963. The EPA restricted firms' ability to pay men and women differently for equal work within the same establishment and was actively enforced at the federal level (Bailey, Helgerman, et al., 2024). Thus, when women worked alongside men in comparable positions, reducing women's wages in response to an increase in their expected fertility costs could also require adjusting the wages of male workers. This constraint should be less relevant in female-dominated jobs, where comparable male workers are rare.

Since we cannot observe the gender composition of individual establishment-position cells, we use the pre-PDA female employment share in an industry as a proxy for the strength of the constraint. Intuitively, establishments in female-dominated industries are more likely to contain positions with few comparable male workers, making the EPA less likely to constrain female wage adjustment. Consistent with this intuition, Table 4 shows a striking difference across industries. In female-dominated industries, where at least 75% of workers are women, wages fall by approximately 0.06 standard deviations while employment in these industries changes little. In the remaining industries, wages are essentially unchanged while employment falls substantially. The fact that the larger wage decline occurs where the employment

response is small makes it less likely that this wage pattern is due to changes in selection into employment.

The industry heterogeneity can also be rationalized through our framework. Viewing the model as describing a single industry, variation across industries can be represented by differences in its primitives, including the scope for substitution between male and female labor. If female-dominated industries offer less scope for substitution toward male labor, then the fertility-cost shock tends to generate relatively less quantity adjustment and more wage adjustment in these industries. Conversely, if more gender-integrated industries offer greater scope for substitution toward male labor, demand for fertile-age female labor is more responsive, producing a larger employment response and a smaller wage response. This is the same qualitative pattern we observe in the data.

This mechanism and the EPA channel are not mutually exclusive. Female-dominated industries may simultaneously face weaker equal-pay constraints and less scope for substitution toward male labor, while more gender-integrated industries may face both stronger wage constraints and greater scope for substitution away from fertile-age women. Both forces would therefore shift adjustment toward wages in the former and toward employment in the latter. We view the evidence as consistent with equal-pay constraints contributing to wage rigidity, potentially reinforced by systematic differences in the scope for substitution across gender inputs across industries.

Job separations upon pregnancy. Dismissal upon pregnancy was common before the PDA and remained frequent afterward, although the reform may have reduced its incidence. As documented in Section 3, dismissing pregnant workers was widespread before the PDA, while post-reform evidence from the PSID shows that involuntary separations among pregnant women remained frequent and substantially more likely than among workers overall (Table 1). Our best proxy for dismissal after pregnancy in the CPS ASEC, observed both before and after the PDA, yields a small and statistically insignificant effect.[31] Appendix Figure A1 further shows that sex-discrimination charges involving discharge increased after the PDA and remained high. Together, this evidence suggests that, while the PDA may have increased retention in some firms, it did not eliminate pregnancy-related dismissals. This pattern is consistent with evidence from the literature that firing costs can discourage hiring without a large reductions in layoffs unless penalties are sufficiently stringent (Behaghel et al., 2008), and that anti-discrimination mandates may fail to eliminate discriminatory practices among "reluctant actors" (Anderson et al., 2024).

[31]Because pregnancy and dismissal are both infrequent in the overall sample, this measure has limited power to detect changes in this margin.

Two points are important for interpreting these findings. First, as discussed in Section 3.1, the PDA increases firms' expected fertility costs through higher dismissal sanctions, even when firms continue dismissing workers upon pregnancy. Second, our employment estimates capture the net effect on fertile-age women, including any increase in employment due to greater retention upon pregnancy. As formalized in Appendix A.2.3, such an increase would partly offset the employment decline induced by greater expected-fertility discrimination, leading us to underestimate the magnitude of the latter effect.

Finally, we find essentially no effect of the PDA on pregnancy rates in either the CPS ASEC or PSID. This rules out the possibility that the employment decline was driven by contemporaneous changes in fertility. However, fertility conditional on employment declines, consistent with women with higher expected fertility becoming less likely to be employed.

6.3 Equilibrium Effects on Other Workers and Aggregate Effects

The PDA reduced men's wages with little change in their employment, while employment and wages of older women remained largely unaffected. able A4 reports these responses in the CPS ASEC and PSID. For men, the wage response is consistent with a contraction in demand for male labor due to technological complementarity with female labor. A response on wages more than employment is expected given the inelastic male labor supply at the time (Elder et al., 2025). For older women, substitution away from fertile-age women can increase labor demand, while the overall contraction in female labor demand works in the opposite direction. The small estimated responses are consistent with these forces approximately offsetting.

Examining employment and wage effects across occupations reveals that the decline among fertile-age women was concentrated in middle-skill jobs, while men's responses varied substantially across occupational groups (Figure 5). The concentration among fertile-age women, particularly in clerical occupations, provides an additional plausibility check given the importance of these jobs in female employment. Among men ages 18–35, wage declines occur primarily outside middle-skill occupations, particularly in high-skill professional and managerial jobs. Among older men, wages decline in low- and middle-skill occupations, while employment falls in high-skill and rises in middle-skill occupations. These patterns are consistent with the mechanisms in our framework: demand for men falls in occupations that are complementary to the ones where female employment contracts, while substitution toward older men can partly offset employment losses in the occupations most affected.

On aggregate, the PDA reduced employment and economic activity, with employment per capita falling by 1.9 percentage points and GDP per capita by approximately 2.5%. Table A5 reports these estimates based on administrative data from the County Business

Patterns (CBP) on state-level employment and establishments.[32]. Establishments per capita also declined by about 2.7%, relative to the pre-PDA mean, while average employment per establishment fell by about 2.0%.

The contraction in employment losses was accompanied by a 2.5% decline in GDP per capita, which is comparable in magnitude to output effects of employment protection and labor regulation documented in previous work (Hopenhayn and Rogerson, 1993; Autor et al., 2007; Bjuggren, 2018). If anything, the decline is smaller than implied by applying aggregate output elasticities with respect to labor (Vollrath, 2024). The smaller output response may partly reflect the concentration of employment losses in clerical occupations, if these jobs have lower marginal contributions to aggregate output than the average job.

6.4 Robustness

We conclude by assessing the robustness of our main employment results. We first examine whether they could be driven by other components of the PDA, then test their sensitivity to alternative samples, specifications, and identifying variation, and finally report additional outcomes and further evidence of heterogeneous effects.

Other components of the PDA. The PDA required equal treatment of pregnancy and comparable temporary disabilities in dimensions other than dismissal as well. In particular, in firms that already provided health insurance or temporary-disability benefits to other eligible workers, the Act could require extending these benefits to pregnant workers. Such expansions can themselves affect employment: they increase the cost of employing fertile-age women, while benefits valued by workers may simultaneously increase labor supply.[33] These components of the PDA have been studied directly by Gruber (1994) for maternity health-insurance coverage and Timpe (2024) for paid maternity leave through temporary-disability insurance. Both find substantial wage adjustment, accompanied by little employment response in Gruber (1994) and a more moderate employment decline in Timpe (2024). These estimates isolate specific benefit expansions that applied only where employers already provided comparable benefits, either voluntarily or under state mandates, whereas our estimates capture the PDA as a whole. We next show that the large aggregate employment decline we document is driven by the Act's more broadly applicable employment-protection provisions rather than by the

[32]CBP data cover aggregate employment for a subset of the workforce. Excluded categories are self-employed, private-household employees, railroad employees, agricultural-production employees, and most government employees.

[33]This is the standard mandated-benefits logic: because workers may value the benefit, part of its cost can be offset by lower equilibrium wages, making mandated benefits potentially efficient relative to cash compensation in some settings (Summers, 1989).

expansion of employer-provided benefits.

To do this, using a strategy similar to Timpe (2024), we exploit pre-PDA differences across industries in the prevalence of employer-provided health benefits, measured in the 1976 Survey of Income and Education. Figure A10 divides industries into terciles of pre-PDA health benefit coverage and estimates the DID with the CPS ASEC separately for each industry group. The employment decline is present even in industries where employer-provided benefits were rare and, if anything, is larger in magnitude in the lower-coverage groups than in industries where these benefits were most common. This is difficult to reconcile with benefit expansions being the primary explanation for the aggregate employment effect: the increase in employment-protection costs applied broadly, whereas the benefit channel should have been substantially weaker precisely in the low-coverage industries where we continue to find large employment declines.

As an additional check, we exploit the fact that Massachusetts was the only PDA-control state that had already mandated employment protection for pregnant workers, through job-protected unpaid leave, without also requiring equal treatment in other dimensions of employment. As a result, the PDA expanded employer-provided health and temporary-disability benefits in Massachusetts in 1978, as it did in PDA-treated states, whereas pregnancy employment protection had already been introduced there in 1972. We therefore replicate our estimates using Massachusetts as the only control state, comparing states that were similarly exposed to the expansion of other benefits in 1978 but differed in whether the PDA newly increased employment protection.[34] The resulting employment effects, reported in Appendix Table A7, are similar to the baseline, although less precise. This further supports the interpretation that the aggregate employment decline is primarily driven by the PDA's employment-protection provisions rather than by benefit expansions.

Identification and specification robustness. A broad set of additional exercises supports the baseline employment result. First, Figure A8 shows that the estimate is not driven by any individual treated or control state. Second, using the PSID, we obtain similar results when we expand the set of admissible treated and control states (Appendix Figure A9 and columns (1)–(2) of Appendix Table A10) and when treatment is defined using respondents' childhood state rather than contemporaneous residence (columns (3)–(4) of Appendix Table A10), reducing concerns about spillovers via selective migration. Third, controlling for contemporaneous

[34]According to the 1976 Survey of Income and Education, employer-provided health insurance was similarly prevalent in Massachusetts and PDA-treated states, at 55.6% and 54.8%, respectively. This similarity supports the use of Massachusetts as a control for isolating the employment-protection component of the PDA, since the two groups of states should have been comparably exposed to the expansion of employer-provided benefits in 1978.

policy changes, such as Head Start availability, employment-at-will exceptions, and unilateral-divorce laws, leaves the results essentially unchanged (Appendix Table A12).

We also obtain similar conclusions from designs that rely on different identifying variation and much larger samples. Using Census data, we do within-state comparisons of fertile-age women with high and low predicted fertility before and after 1978 and find that employment falls disproportionately among women with greater predicted fertility (Appendix Table A11). The same pattern is found for early-treated states before and after their own state policies comparable to the PDA.

Finally, Appendix Table A6 reports triple-difference specifications using men or older women as within-state comparison groups. These estimates are similar to the baseline and have the advantage of flexibly absorbing state-specific shocks. We do not use them as our preferred specification, however, because their interpretation requires men or older women to be unaffected by the PDA. As Section 6.3 shows, this restriction is not implied by our framework: changes in the employment of fertile-age women can affect both groups through equilibrium interactions across labor inputs. The triple-difference estimates therefore rely on a stronger identifying assumption than our baseline design, and we view them as a useful robustness exercise rather than as an alternative preferred specification.

Additional outcomes and heterogeneity. Appendix Table A8 reports a number of additional outcomes. The PDA increases the probability that employed fertile-age women work part time and reduces weekly hours, while having little effect on the number of weeks worked conditional on employment. These results suggest that the PDA affected the intensive margin of employment as well, but the extensive-margin employment response remains quantitatively larger. An alternative explanation to the employment decline is that pregnancy protections increased expected job tenure and returns to education, inducing women to remain in school rather than work. However, we find no increase in college completion, which instead declines slightly.

We additionally report heterogeneity by race, education and marital status in Appendix Table A9. In particular, employment effects are larger among less-educated women. This does not imply that the cost of pregnancy conditional on childbirth is greater in low-skill jobs. Expected fertility costs depend both on the consequences of a pregnancy for the firm and on the probability that pregnancy occurs, and fertility rates in our period were substantially higher among less-educated women. Differences in labor-supply elasticities, substitution patterns, or firms' pre-PDA firing behavior may also contribute to the heterogeneous equilibrium response.[35] Employment declines are also larger among married and Black women, consistent

[35]Another possible interpretation is that expected-fertility discrimination was already more important in

with higher fertility rates among these groups during the period.

7 Quantification of Expected-Fertility Discrimination

Having established that the PDA increased expected-fertility discrimination, we now quantify the contribution of this increase to the observed gender employment gap and evaluate the conditions for it to provide a lower bound on the total contribution of this form of discrimination.

We use the unconditional gender employment gap as the benchmark for three reasons. First, our object of interest is the contribution of expected-fertility discrimination to the gender disparities actually observed in the population. The raw gender gap is therefore the natural denominator for asking how much of that disparity this mechanism accounts for.[36] Second, we focus on employment because it is observed for the entire population of interest and because the PDA generates substantial effects on employment and hiring in our setting. Hourly wages, by contrast, are observed only among employed workers, and the PDA itself changes this selection. Third, the unconditional gap is also preferable given the role of observables in expected-fertility discrimination. Traditionally, the literature has interpreted unexplained gaps remaining after conditioning on productivity-related characteristics as proxies for discrimination (Oaxaca, 1973; Blinder, 1973; Altonji and Blank, 1999). More recent work has emphasized that such characteristics may themselves reflect pre-market or systemic discrimination (Blau and Kahn, 2017; Lundberg, 2022; Bohren et al., 2025).[37] Expected-fertility discrimination raises a distinct issue: observables may be *used* to discriminate. Characteristics such as age and education help predict women's future fertility and can therefore determine the extent of differential treatment they face. Variation across these characteristics is therefore part of the mechanism through which expected-fertility discrimination operates, making a benchmark gap that retains this variation preferable for our purpose.

The gender employment gap among fertile-age individuals was 17.4 p.p. in PDA-treated states after 1978 (Fig. 6). We focus on these states and time period since our empirical strategy identifies the presence of discrimination with the PDA enacted, in affected states. Computing the difference-in-differences for the gap, we estimate that it would have been

high-skill jobs before the PDA, leaving less scope for the reform to increase it further. Such a mechanism would require nonlinearities or heterogeneity beyond the baseline model, however, so we do not rely on it in interpreting the results.

[36] A gap conditional on worker characteristics would instead answer a different question: how much expected-fertility discrimination contributes to the gender gap after holding the distribution of those characteristics fixed.

[37] Related concerns arise when these characteristics are themselves chosen or reported by the decision maker (Fryer, 2019; Durlauf and Heckman, 2020).

4.7 p.p. smaller in the counterfactual absent the PDA. In relative terms, this implies that the PDA-induced increase in expected-fertility discrimination accounted for 27% of the gap observed after the PDA.

We next explore heterogeneity by motherhood status, which we expect to be an important margin for expected-fertility discrimination. To the extent that motherhood is observable or inferable, employers considering whether to hire current mothers know that the transition to motherhood has already occurred and may therefore have more information about subsequent labor-force attachment and productivity. For non-mothers, by contrast, a future pregnancy may entail not only the short-run costs associated with leave, temporary replacement, and accommodations, but also greater uncertainty about how the transition to motherhood will affect longer-run labor supply. Expected fertility may therefore play a more important role in shaping employers' expectations when considering whether to hire non-mothers.

Consistent with this hypothesis, expected-fertility discrimination accounts for a substantially larger share of the gender employment gap among non-mothers. Replicating the calculation separately for respondents aged 18 to 35 with and without children, we find that the PDA-induced increase in expected-fertility discrimination accounts for 54% of the gap among fertile-age non-mothers, compared with 23% among mothers (Figure 6). The share is therefore substantially larger among non-mothers even though their underlying gender employment gap is smaller.

As stated in Section 2, these estimates capture only the increase in expected-fertility discrimination induced by the PDA. Even before the reform, dismissing pregnant workers could entail costs, including replacement and training costs, so fertility costs may already have been positive. Denoting pre- and post-PDA fertility costs by C^{NoPDA} and C^{PDA}, respectively, our estimate captures $\mathcal{S}(C^{\text{PDA}}, C^{\text{NoPDA}}) = 27\%$, whereas the total contribution of expected-fertility discrimination to the gender employment gap is $\mathcal{S}(C^{\text{PDA}}, 0)$.

The PDA-induced contribution provides a lower bound on the total contribution if and only if $\Gamma(C^{\text{NoPDA}}) \geq \Gamma(0)$, given the positive post-PDA gender employment gap. That is, the gap at pre-existing expected-fertility discrimination must be weakly larger than the gap in the counterfactual without discrimination. A sufficient condition is that $\Gamma(C)$ be weakly increasing over $[0, C^{\text{NoPDA}}]$.

The model alone does not guarantee this monotonicity condition. The model predicts that fertile-age women's employment declines as C increases, but men's employment may also decline through equilibrium spillovers. One feature of our setting nevertheless favors a widening gap: women's labor supply was substantially more elastic than men's around 1980, making employment adjustments more pronounced among women and wage adjustments more pronounced among men (Elder et al., 2025). Because this difference alone does not guarantee

the required condition, we assess it through a calibration of the equilibrium model.[38]

To calibrate the model, we need to specify a functional form for the production and labor supply functions, which we assume to be, respectively, CES with constant returns to scale following Card and Lemieux (2001), and isoelastic.[39] For the relevant elasticities and substitution parameters, we use estimates from the literature (Card and Lemieux, 2001; Acemoglu, Autor, et al., 2004; Attanasio et al., 2018) and calibrate baseline equilibrium quantities to the pre-PDA data. Importantly, actual fertility costs C^{NoPDA} and C^{PDA} are not identified. Hence, we consider a wide range of possible values and retain those whose predicted employment responses to the PDA are not rejected given the empirical estimates. Details on this exercise are reported in Appendix A.3.

Across the empirically plausible parameterizations we consider, the calibrated model satisfies the required monotonicity condition: tracing the calibrated equilibrium from $C = 0$ to $\phi = \phi^{\text{NoPDA}}$, we find that $\Gamma(C)$ is weakly increasing throughout this interval.[40]

The calibration thus supports interpreting the 27% contribution identified through the PDA as a lower bound on the total contribution of expected-fertility discrimination. Under this interpretation, eliminating expected-fertility discrimination would have reduced the post-PDA gender employment gap among fertile-age individuals by at least 27%.

8 Conclusion

This paper studies discrimination against women based on employers' expectations about their future fertility. We exploit the Pregnancy Discrimination Act of 1978, which increased firms' expected costs associated with future childbirth, and compare states newly affected by the federal reform with states that had already adopted comparable protections. We find that the PDA reduced employment of fertile-age women by 6.6%, primarily through lower hiring. The decline is concentrated among women with higher expected fertility, providing evidence that employers act on the anticipated costs of future childbirth. Quantitatively, our estimates imply that expected-fertility discrimination accounts for at least 27% of the gender employment gap among fertile-age individuals in our setting.

Our estimates come from the U.S. labor market of the late 1970s and early 1980s, but the

[38] For the empirical and calibration analyses, we express the gender gap in employment-to-population ratios. The lower-bound condition applies also to this employment-rate gap, which we formalize in Appendix A.2.2.

[39] This standard specification of the production function makes female and male effective labor Edgeworth complements, so the contraction in female labor also reduces male employment. Because lower male employment offsets the widening of the gender employment gap generated by the decline in female employment, this male-employment response works against the monotonicity condition we seek to establish.

[40] In Appendix A.2.4, when we discuss other channels through which the PDA might have affected labor market outcomes, we argue that the lower bound interpretation holds true also in those cases.

mechanism they identify remains relevant today. Employers continue to face costs associated with childbirth, while fertility rates are broadly similar in magnitude to those prevailing around the passage of the PDA.[41] The incidence of these costs, however, has evolved in opposing directions. Stronger employment protections, leave policies, and benefits around childbirth plausibly increased employers' exposure to fertility-related costs, while greater labor-force attachment after childbirth and changes in workplace organization might have reduced the disruptions associated with fertility. Contemporary evidence is consistent with employers continuing to respond to expected fertility (e.g., Becker et al., 2019; Bamieh and Ziegler, 2023; He et al., 2023; Landais et al., 2026).

The margins through which this discrimination operates may also have changed. In our historical setting, adjustment occurred primarily through employment and hiring, while wage adjustment was relatively limited. In modern labor markets, stronger enforcement of discrimination in hiring and pay may shift differential treatment toward less observable margins, including promotions, job assignments, or career progression. Moreover, the relative importance of expected-fertility discrimination need not have fallen even if its absolute magnitude has not increased. Outright discrimination in the labor market has diminished substantially since the 1970s (Cortes et al., 2026), while educational attainment has shifted in women's favor. Expected fertility may therefore account for a larger share of the gender disparities that remain.

More broadly, our results highlight an important consideration for policies surrounding childbirth. Protecting workers once pregnancy or childbirth occurs can affect firms' decisions before fertility is realized (Thomas, 2020). When such policies increase the net costs of childbirth borne by employers, and discrimination on ex-ante margins such as hiring, pay, or promotion is difficult to detect or enforce, they can strengthen incentives for expected-fertility discrimination. Policy design should therefore consider not only the protections provided after childbirth, but also the incidence of the associated costs. If the objective extends beyond protecting current mothers to reducing gender disparities more broadly, limiting employers' exposure to childbirth-related costs can help prevent protections around realized fertility from generating differential treatment based on expected fertility.

[41] The average number of births per woman was 1.76 in 1978. From then until 2017, it remained above this level. See here.

References

Acemoglu, Daron and Joshua D Angrist (2001). "Consequences of employment protection? The case of the Americans with Disabilities Act". *Journal of Political Economy* 109.5, pp. 915–957.

Acemoglu, Daron, David H Autor, and David Lyle (2004). "Women, war, and wages: The effect of female labor supply on the wage structure at midcentury". *Journal of political Economy* 112.3, pp. 497–551.

Agan, Amanda and Sonja Starr (2018). "Ban the box, criminal records, and racial discrimination: A field experiment". *The Quarterly Journal of Economics* 133.1, pp. 191–235.

Agarwal, Sumit, Keyang Li, Yu Qin, Jing Wu, and Jubo Yan (2024). "The impact of fertility relaxation on the gender wage gap". *Journal of Human Resources*.

Altonji, Joseph G and Rebecca M Blank (1999). "Race and gender in the labor market". *Handbook of labor economics* 3, pp. 3143–3259.

Anderson, D Mark, Kerwin Kofi Charles, and Daniel I Rees (2024). "Imposing policy on reluctant actors: The hospital desegregation campaign and Black postneonatal mortality in the Deep South". *Review of Economics and Statistics*, pp. 1–46.

Angrist, Joshua D and Jörn-Steffen Pischke (2009). *Mostly harmless econometrics: An empiricist's companion*. Princeton university press.

Arefeva, Alina, Kay Jowers, Qihui Hu, and Christopher Timmins (2024). *Enforcement Risk and Screening in Rental Markets: Evidence from Eviction Moratoria*. Tech. rep. National Bureau of Economic Research, Inc.

Arnold, David, Will Dobbie, and Crystal S Yang (2018). "Racial bias in bail decisions". *The Quarterly Journal of Economics* 133.4, pp. 1885–1932.

Attanasio, Orazio, Peter Levell, Hamish Low, and Virginia Sánchez-Marcos (2018). "Aggregating elasticities: intensive and extensive margins of women's labor supply". *Econometrica* 86.6, pp. 2049–2082.

Autor, David H, William R Kerr, and Adriana D Kugler (2007). "Does employment protection reduce productivity? Evidence from US states". *The Economic Journal* 117.521, F189–F217.

Azmat, Ghazala and Barbara Petrongolo (2014). "Gender and the labor market: What have we learned from field and lab experiments?" *Labour economics* 30, pp. 32–40.

Bailey, Martha J, John DiNardo, and Bryan A Stuart (2021). "The economic impact of a high national minimum wage: Evidence from the 1966 fair labor standards act". *Journal of labor economics* 39.S2, S329–S367.

Bailey, Martha J, Thomas Helgerman, and Bryan A Stuart (Feb. 2024). "How the 1963 Equal Pay Act and 1964 Civil Rights Act Shaped the Gender Gap in Pay*". *The Quarterly Journal of Economics*, qjae006. ISSN: 0033-5533. DOI: 10.1093/qje/qjae006.

Bamieh, Omar and Lennart Ziegler (2023). "Gender-age differences in hiring rates and prospective wages—Evidence from job referrals to unemployed workers". *Labour Economics* 83, p. 102395.

Becker, Sascha O, Ana Fernandes, and Doris Weichselbaumer (2019). "Discrimination in hiring based on potential and realized fertility: Evidence from a large-scale field experiment". *Labour Economics* 59, pp. 139–152.

Behaghel, Luc, Bruno Crépon, and Béatrice Sédillot (2008). "The perverse effects of partial employment protection reform: The case of French older workers". *Journal of Public Economics* 92.3-4, pp. 696–721.

Bell, Robert M and Daniel F McCaffrey (2002). "Bias reduction in standard errors for linear regression with multi-stage samples". *Survey Methodology* 28.2, pp. 169–182.

Berglund, Patricia A and Steven G Heeringa (2015). "Panel Study of Income Dynamics PSID Cross-sectional Individual Weights, 1997-2013". *Ann Arbor, MI: Institute for Social Research*.

Bertrand, Marianne and Sendhil Mullainathan (2004). "Are Emily and Greg more employable than Lakisha and Jamal? A field experiment on labor market discrimination". *American economic review* 94.4, pp. 991–1013.

Bjuggren, Carl Magnus (2018). "Employment protection and labor productivity". *Journal of Public Economics* 157, pp. 138–157.

Blau, Francine D (2025). "Gender Inequality in the Labor Market: Continuing Progress?" *ILR Review* 78.2, pp. 275–303.

Blau, Francine D and Lawrence M Kahn (2017). "The gender wage gap: Extent, trends, and explanations". *Journal of economic literature* 55.3, pp. 789–865.

Blinder, Alan S. (1973). "Wage Discrimination: Reduced Form and Structural Estimates". *The Journal of Human Resources* 8.4, pp. 436–455. (Visited on 08/13/2026).

Bohren, J Aislinn, Peter Hull, and Alex Imas (2025). "Systemic discrimination: Theory and measurement". *The Quarterly Journal of Economics* 140.3, pp. 1743–1799.

Brock, J Michelle and Ralph De Haas (2023). "Discriminatory lending: Evidence from bankers in the lab". *American Economic Journal: Applied Economics* 15.2, pp. 31–68.

Canay, Ivan A, Andres Santos, and Azeem M Shaikh (2021). "The wild bootstrap with a "small" number of "large" clusters". *Review of Economics and Statistics* 103.2, pp. 346–363.

Card, David and Thomas Lemieux (2001). "Can falling supply explain the rising return to college for younger men? A cohort-based analysis". *The quarterly journal of economics* 116.2, pp. 705–746.

Cortes, Patricia, Jisoo Hwang, Jessica Pan, and Uta Schönberg (2026). *Gender norms and the labor market*. Tech. rep. National Bureau of Economic Research.

Cullen, Zoë, Will Dobbie, and Mitchell Hoffman (2023). "Increasing the demand for workers with a criminal record". *The Quarterly Journal of Economics* 138.1, pp. 103–150.

Dobbie, Will, Andres Liberman, Daniel Paravisini, and Vikram Pathania (2021). "Measuring bias in consumer lending". *The Review of Economic Studies* 88.6, pp. 2799–2832.

Doleac, Jennifer L and Benjamin Hansen (2020). "The unintended consequences of "ban the box": Statistical discrimination and employment outcomes when criminal histories are hidden". *Journal of Labor Economics* 38.2, pp. 321–374.

Dowd, Nancy E (1985). "Maternity Leave: Taking Sex Differences into Account". *Fordham L. Rev.* 54, p. 699.

Durlauf, Steven N and James J Heckman (2020). "An empirical analysis of racial differences in police use of force: A comment". *Journal of Political Economy* 128.10, pp. 3998–4002.

Edwards, Mark Evan (1996). "Pregnancy discrimination litigation: Legal erosion of capitalist ideology under equal employment opportunity law". *Social Forces* 75.1, pp. 247–268.

Elder, Todd, Steven J Haider, and Cody Orr (2025). "The evolution of the wage elasticity of labor supply over time". *Journal of Human Resources*.

Ewens, Michael, Bryan Tomlin, and Liang Choon Wang (2014). "Statistical discrimination or prejudice? A large sample field experiment". *Review of Economics and Statistics* 96.1, pp. 119–134.

Fernández-Kranz, Daniel and Núria Rodríguez-Planas (2021). "Too family friendly? The consequences of parent part-time working rights". *Journal of Public Economics* 197, p. 104407.

Flood, Sarah, Renae Rodgers, José Pacas, Devon Kristiansen, and Ben Klaas (2022). "Extending current population survey linkages: Obstacles and solutions for linking monthly data from 1976 to 1988". *Journal of economic and social measurement* 46.1, pp. 1–28.

Folke, Olle and Johanna Rickne (2026). *Misunderstandings of the Child Penalty Graph*. Tech. rep. ROCKWOOL Foundation Berlin (RFBerlin).

Fryer, Roland G Jr (2019). "An empirical analysis of racial differences in police use of force". *Journal of Political Economy* 127.3, pp. 1210–1261.

Gallen, Yana (2023). "Motherhood and the gender productivity gap". *Journal of the European Economic Association*.

Gardin, Susan Kelemen and Gary A Richwald (1986). "Pregnancy and employment leave: Legal precedents and future policy". *Journal of Public Health Policy* 7, pp. 458–469.

Gentile Passaro, Diego, Fuhito Kojima, and Bobak Pakzad-Hurson (2026). "Equal Pay for Similar Work". *American Economic Review* 116.3, pp. 977–1013.

Ghosh, Pulak, Stephanie Hao, Lisa Ho, Garima Sharma, and Shreya Tandon (2025). "The Effects of Mandated Maternity Leave on Labor Market Outcomes in India". *Unpublished manuscript*.

Ginja, Rita, Arizo Karimi, and Pengpeng Xiao (2023). "Employer responses to family leave programs". *American Economic Journal: Applied Economics* 15.1, pp. 107–135.

Goldin, Claudia (1990). "Understanding the gender gap: An economic history of American women".

— (2023). *Why women won*. Tech. rep. National Bureau of Economic Research.

Goldin, Claudia and Cecilia Rouse (2000). "Orchestrating impartiality: The impact of "blind" auditions on female musicians". *American economic review* 90.4, pp. 715–741.

Gruber, Jonathan (1994). "The incidence of mandated maternity benefits". *The American economic review*, pp. 622–641.

Habig, Jill E (2007). "Defining the protected class: Who qualifies for protection under the Pregnancy Discrimination Act". *Yale LJ* 117, p. 1215.

Hagemann, Andreas (2019). "Placebo inference on treatment effects when the number of clusters is small". *Journal of Econometrics* 213.1, pp. 190–209.

He, Haoran, Sherry Xin Li, and Yuling Han (2023). "Labor market discrimination against family responsibilities: A correspondence study with policy change in China". *Journal of Labor Economics* 41.2, pp. 361–387.

Hopenhayn, Hugo and Richard Rogerson (1993). "Job turnover and policy evaluation: A general equilibrium analysis". *Journal of political Economy* 101.5, pp. 915–938.

Hsieh, Chang-Tai, Erik Hurst, Charles I Jones, and Peter J Klenow (2019). "The allocation of talent and us economic growth". *Econometrica* 87.5, pp. 1439–1474.

Katz, Lawrence F and Robert A Margo (2014). "Technical change and the relative demand for skilled labor: The united states in historical perspective". *Human capital in history: The American record*. University of Chicago Press, pp. 15–57.

Kleven, Henrik, Camille Landais, and Gabriel Leite-Mariante (2024). "The child penalty atlas". *Review of Economic Studies*, rdae104.

Landais, Camille, Petter Lundborg, Tatiana Pazem, and Johan Vikstrom (2026). "Gender Without Children".

Lang, Kevin and Ariella Kahn-Lang Spitzer (May 2020). "Race Discrimination: An Economic Perspective". *Journal of Economic Perspectives* 34.2, pp. 68–89. DOI: 10.1257/jep.34.2.68. URL: https://www.aeaweb.org/articles?id=10.1257/jep.34.2.68.

Li, Qingxiao and Di Xiao (2025). "Fertility discrimination in the Chinese labor market: Evidence from a correspondence study and an employer survey". *Labour Economics* 92, p. 102668.

Lundberg, Shelly (2022). "Gender economics and the meaning of discrimination". *AEA Papers and Proceedings*. Vol. 112. American Economic Association 2014 Broadway, Suite 305, Nashville, TN 37203, pp. 588–591.

MacKinnon, James G, Morten Ørregaard Nielsen, and Matthew D Webb (2023). "Cluster-robust inference: A guide to empirical practice". *Journal of Econometrics* 232.2, pp. 272–299.

Maloney, Molly and David Neumark (2025). *Does the Gender Wage Gap Actually Reflect Taste Discrimination Against Women?* Tech. rep. National Bureau of Economic Research.

Marchingiglio, Riccardo and Michael Poyker (2026). "The economics of gender-specific minimum wage legislation". *Journal of Labor Economics* 44.3, pp. 000–000.

Neumark, David and Wendy A Stock (2006). "The labor market effects of sex and race discrimination laws". *Economic Inquiry* 44.3, pp. 385–419.

O'Brien, Christine Neylon and Gerald A Madek (1988). "Pregnancy discrimination and maternity leave laws". *Dick. L. Rev.* 93, p. 311.

Oaxaca, Ronald (1973). "Male-female wage differentials in urban labor markets". *International economic review* 14.3, pp. 693–709.

Panel Study of Income Dynamics (2026). *Public-use dataset*. Ann Arbor, MI.

Pedriana, Nicholas (2009). "Discrimination by Definition: The Historical and Legal Paths to the Pregnancy Discrimination Act of 1978". *Yale JL & Feminism* 21, p. 1.

Pfeffer, Fabian T., Davis Daumler, and Esther Friedman (2025). *PSID-SHELF, 1968–2021: The PSID's Social, Health, and Economic Longitudinal File (PSID-SHELF), Beta Release*. Ann Arbor, MI. DOI: 10.3886/E194322V2.

Pustejovsky, James E and Elizabeth Tipton (2018). "Small-sample methods for cluster-robust variance estimation and hypothesis testing in fixed effects models". *Journal of Business & Economic Statistics* 36.4, pp. 672–683.

Ramanathan, Kumar (2021). "From civil rights to social policy: The political development of family and medical leave policy". *Studies in American Political Development* 35.2, pp. 173–193.

Remmers, Cynthia L (1989). "Pregnancy discrimination and parental leave". *Indus. Rel. LJ* 11, p. 377.

Rose, Evan K (2023). "A constructivist perspective on empirical discrimination research". *Journal of Economic Literature* 61.3, pp. 906–923.

Roth, Jonathan, Pedro HC Sant'Anna, Alyssa Bilinski, and John Poe (2023). "What's trending in difference-in-differences? A synthesis of the recent econometrics literature". *Journal of econometrics* 235.2, pp. 2218–2244.

Siegel, Reva B (1985). "Employment equality under the Pregnancy Discrimination Act of 1978". *The Yale Law Journal* 94.4, pp. 929–956.

Stucke, Adela (1945). "Notes on Compulsory Sickness Insurance Legislation in the States, 1939-44". *Public Health Reports (1896-1970)*, pp. 1551–1564.

Summers, Lawrence H (1989). "Some simple economics of mandated benefits". *The American Economic Review* 79.2, pp. 177–183.

Sun, Ge (2025). "Expected Fertility, Labor Market Contracts, and the Gender Wage Gap".

Thomas, Mallika (2020). "The impact of mandated maternity leave policies on the gender gap in promotions: examining the role of employer-based discrimination". *Available at SSRN 3729663*.

Timpe, Brenden (2024). "The labor market impacts of America's first paid maternity leave policy". *Journal of Public Economics* 231, p. 105067.

US Bureau of the Census (1992). *Survey of Income and Education, 1976: Rectangular File*. DOI: 10.3886/ICPSR07919.v1.

Vollrath, Dietrich (Oct. 2024). "The Elasticity of Aggregate Output with Respect to Capital and Labor". *American Economic Journal: Macroeconomics* 16.4, pp. 470–504.

Weissmann, Andrew (1983). "Sexual equality under the pregnancy discrimination act". *Colum. L. Rev.* 83, p. 690.

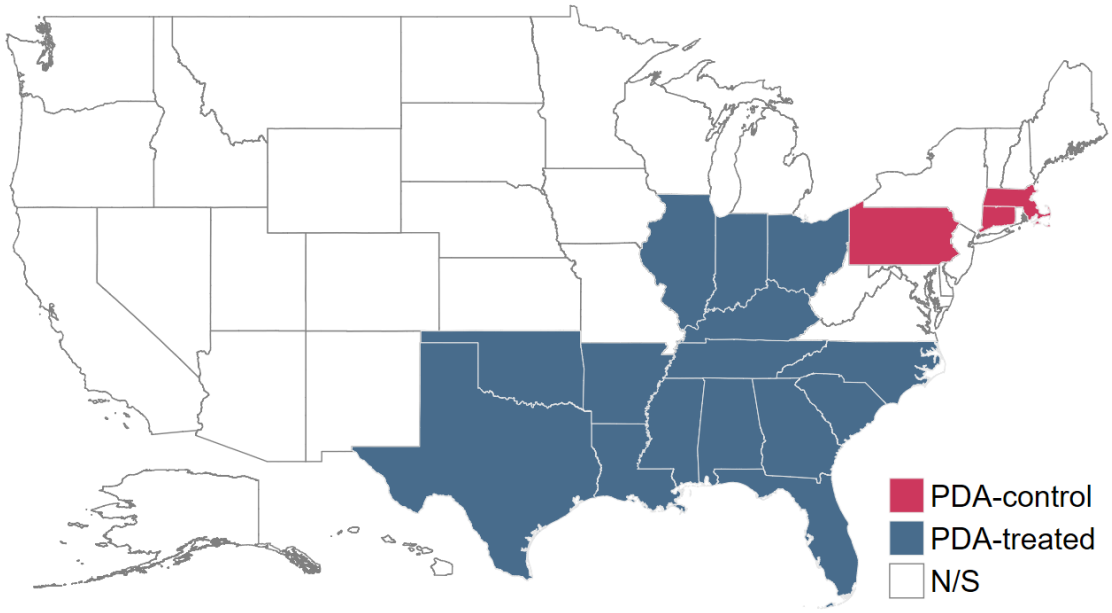
Wolkinson, Benjamin W and Dennis H Liberson (1982). "The Arbitration of Sex Discrimination Grievances." *Arbitration Journal* 37.2.

Wozniak, Abigail (2015). "Discrimination and the effects of drug testing on black employment". *Review of Economics and Statistics* 97.3, pp. 548–566.

Zabalza, Antoni and Zafiris Tzannatos (1985). "The effect of Britain's anti-discriminatory legislation on relative pay and employment". *The Economic Journal* 95.379, pp. 679–699.

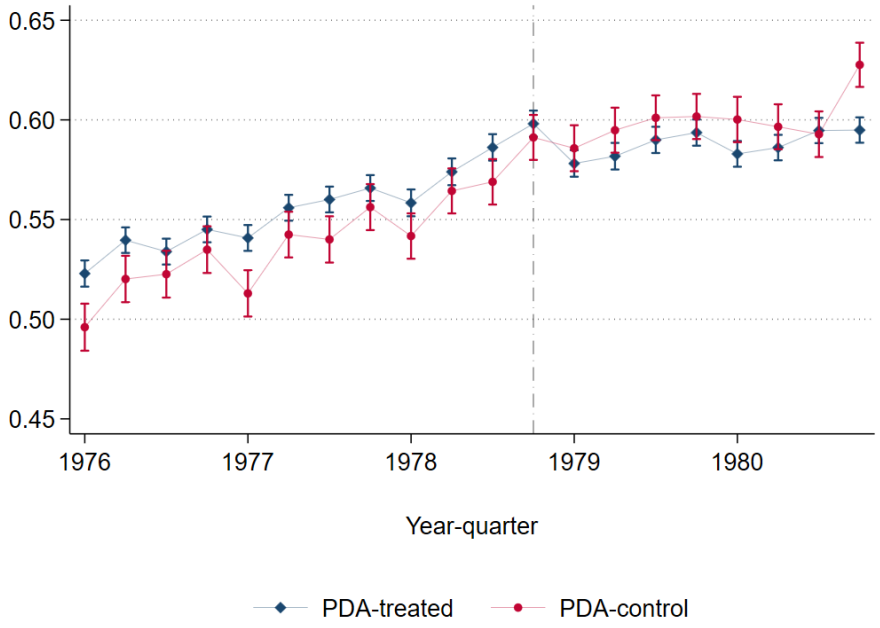
Main Figures & Tables

Figure 1: PDA-treated and PDA-control States, Main Sample



Notes: This map displays the set of PDA-treated and PDA-control states used in the main analyses. N/S stands for 'not in sample'. The full set of states used in robustness analyses with the PSID are shown in Figure A9.

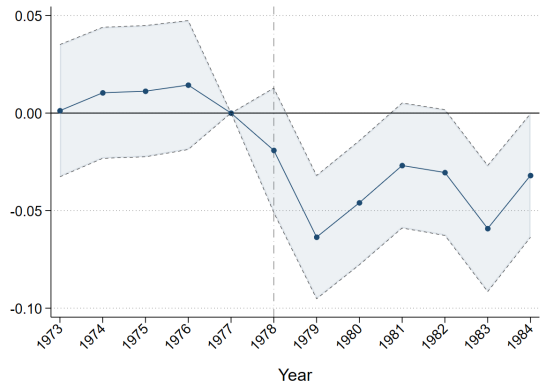
Figure 2: Raw Trends in Employment



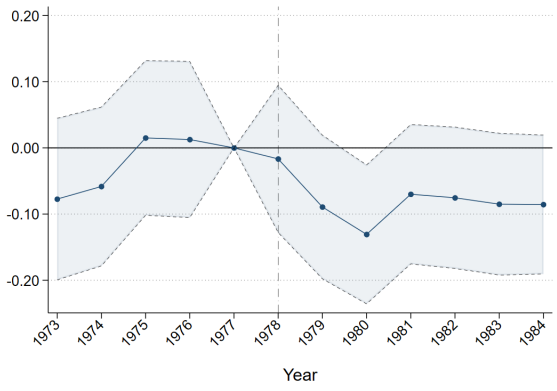
Notes: The sample is restricted to women of age 18 to 35 in PDA-treated and PDA-control states between 1976 and 1980, from the basic monthly CPS. Plotted are the average values of the dummy for being employed for each group of states in the year indicated on the x-axis. Estimates are computed using CPS individual sample weights. 95% confidence intervals are based on heteroskedasticity robust standard errors.

Figure 3: Event-Study Estimates: Employment

(a) CPS ASEC



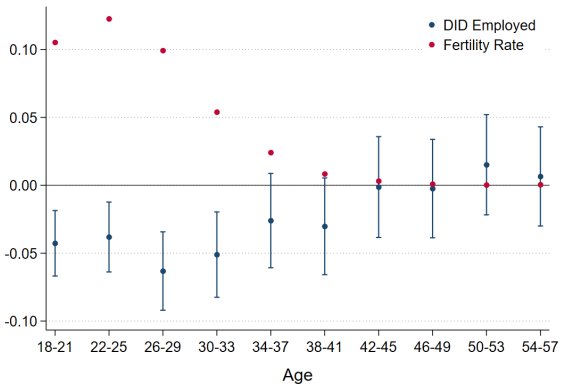
(b) PSID



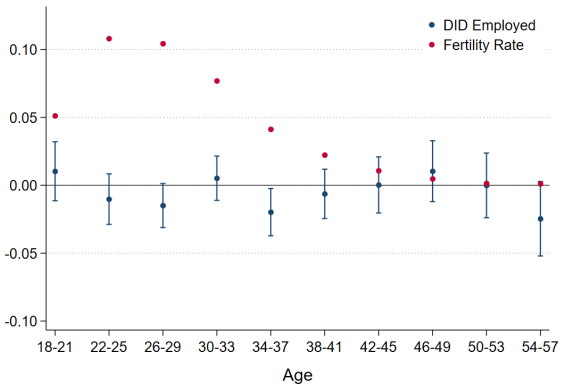
Notes: The sample consists of all female respondents of age 18 to 35 in PDA-treated and PDA-control states, between 1973 and 1984 from CPS ASEC, in panel (a), and PSID, in panel (b). Estimates are computed using each survey's individual sample weights. 95% confidence intervals are based on heteroskedasticity robust standard errors.

Figure 4: Fertility Rate and Effect on Employment by Age Group

(a) Women



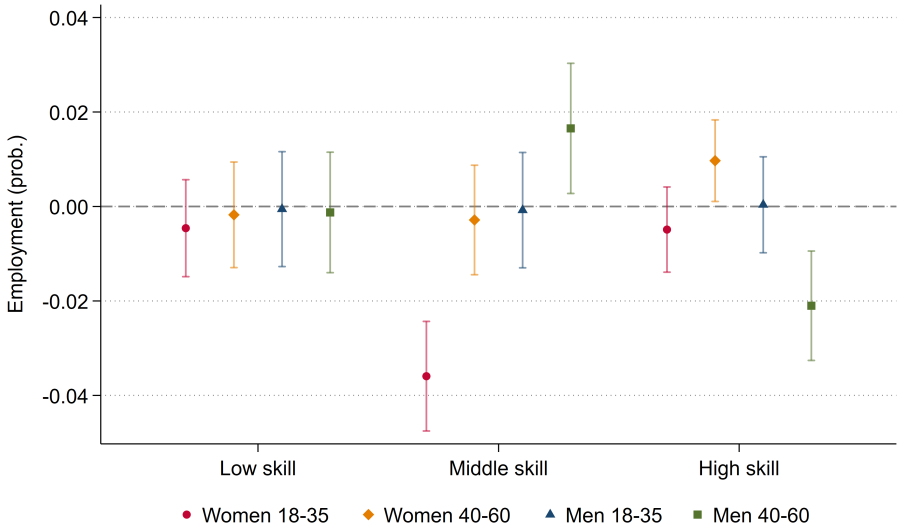
(b) Men



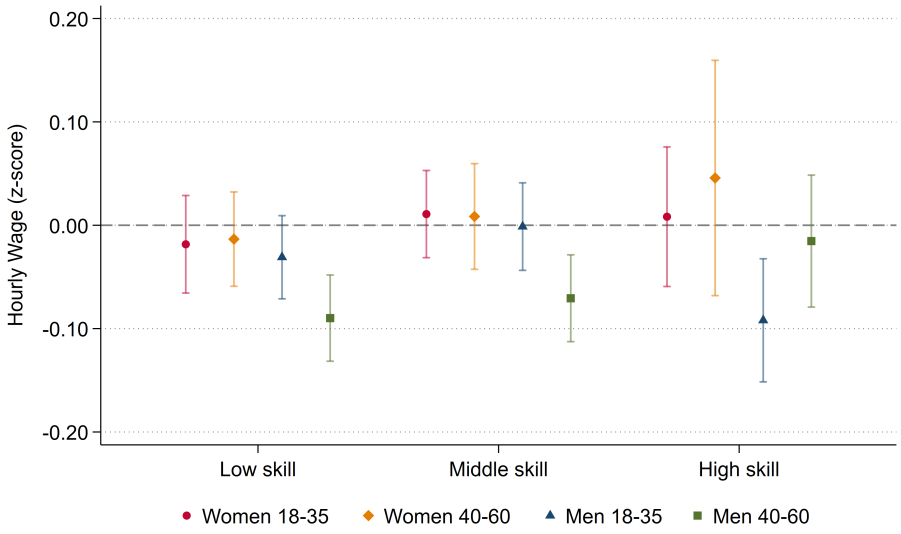
Notes: The sample is restricted to individuals of age 18 to 60 in PDA-treated and PDA-control states between 1973 and 1984, from the CPS ASEC. Estimates are computed based on specification (7), using ASEC individual sample weights, separately within each age group. Given the narrow age groups, we only include age and age squared among the covariates, as they are almost perfectly collinear in some age groups. Employed is a dummy equal to 1 if the respondent is employed at some point in the year. The fertility rate is the average share of respondents having a child at that age. 95% confidence intervals are based on heteroskedasticity robust standard errors.

Figure 5: Effects of the PDA by Occupational Skill Level and Demographic Group

(a) Employment

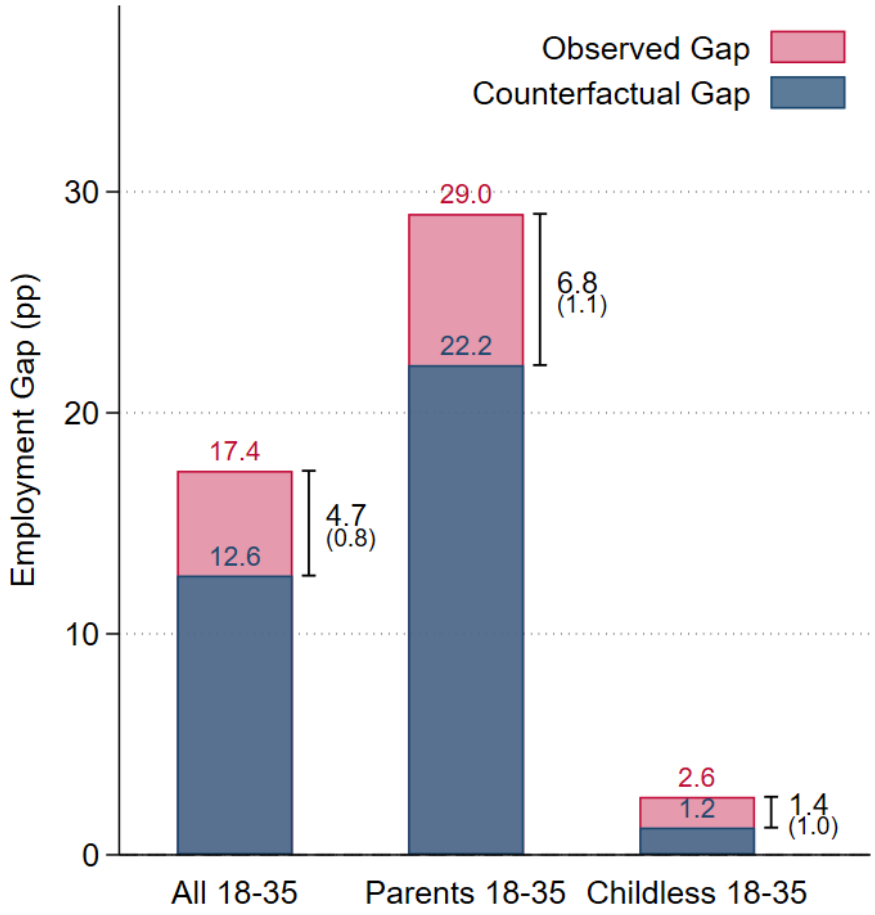


(b) Hourly Wages



Notes: The sample is restricted to the demographic groups indicated in the legend, in PDA-treated and PDA-control states between 1973 and 1984, from the CPS ASEC. Occupations codes (IPUMS 1950 classification) are grouped by skill level following the Katz and Margo (2014). Low skill: Operatives, service workers, and farm and nonfarm laborers. Middle skill: Farmers and farm managers, clerical workers, sales workers, craftsmen and foremen. High skill: Professional and technical workers, managers, officials, and proprietors. Panel (a) uses as outcome a dummy equal to one if the respondent is employed in an occupation with the skill-level indicated at the bottom of the panel and zero otherwise. Panel (b) further restricts the sample to respondents employed in the occupation group indicated at the bottom of the panel, and estimate specification (7) using hourly wage as outcome. Hourly wage is expressed as a z-score within each group. It is constructed using annual pre-tax wage income, number of weeks employed and usual weekly working hours. Estimates are computed using ASEC individual sample weights and include the fixed effects indicated in the table. Standard errors are robust to heteroskedasticity.

Figure 6: Effects on Gender Employment Gaps



Notes: The sample is restricted to women of age 18 to 35. The gender employment gap (in pink) was estimated using CPS ASEC data, restricted to respondents in PDA-treated states between 1979 and 1984. It was estimated regressing a dummy for being employed on a dummy for gender. The counterfactual gap absent the PDA (in blue) was estimated as the observed gap minus the effect of the PDA on the gap, obtained using specification (7), without covariates. The difference between the observed and counterfactual gaps, which is the estimated effect of the PDA on the gap is reported next to the bars, with the heteroskedasticity robust standard errors in parentheses. In the second bar, both gaps were estimated restricting the sample to respondents with at least one child already born at the time of the survey. In the third bar, to those with no child born at the time of the survey.

Table 1: Summary Statistics, CPS ASEC and PSID data

Variable	Mean	Std. Dev.	Min.	Max.	Obs.	Source
Panel A. Women 18–35						
High school	0.813	0.390	0	1	115,217	CPS ASEC
College	0.351	0.477	0	1	115,217	CPS ASEC
White	0.834	0.372	0	1	115,217	CPS ASEC
Married	0.614	0.487	0	1	115,217	CPS ASEC
Number of children	1.24	1.28	0	9	18,984	PSID
Employed	0.719	0.449	0	1	115,217	CPS ASEC
Employed	0.609	0.488	0	1	14,575	PSID
Hourly wage	4.86	3.73	0.00	100.00	68,374	CPS ASEC
Hourly wage	5.16	3.99	0.00	99.99	12,348	PSID
Pregnant	0.087	0.282	0	1	115,217	CPS ASEC
Pregnant	0.094	0.291	0	1	18,989	PSID
Employed while pregnant	0.567	0.495	0	1	10,165	CPS ASEC
Employed while pregnant	0.564	0.496	0	1	4,337	PSID
Job separation after pregnancy	0.421	0.494	0	1	916	PSID
Quit after pregnancy*	0.316	0.465	0	1	884	PSID
Fired/laid off after pregnancy	0.038	0.192	0	1	884	PSID
Panel B. Women 40–60						
High school	0.642	0.480	0	1	90,396	CPS ASEC
College	0.201	0.401	0	1	90,396	CPS ASEC
White	0.869	0.338	0	1	90,396	CPS ASEC
Married	0.742	0.438	0	1	90,396	CPS ASEC
Number of children	3.15	2.35	0	15	9,255	PSID
Employed	0.599	0.490	0	1	90,396	CPS ASEC
Employed	0.577	0.494	0	1	7,754	PSID
Hourly wage	5.21	3.88	0.00	100.00	41,311	CPS ASEC
Hourly wage	5.59	5.12	0.00	99.99	6,289	PSID
Panel C. Men 18–60						
High school	0.739	0.439	0	1	204,212	CPS ASEC
College	0.358	0.479	0	1	204,212	CPS ASEC
White	0.869	0.337	0	1	204,212	CPS ASEC
Married	0.685	0.465	0	1	204,212	CPS ASEC
Number of children	1.85	1.92	0	15	23,733	PSID
Employed	0.905	0.293	0	1	204,212	CPS ASEC
Employed	0.864	0.343	0	1	26,985	PSID
Hourly wage	7.83	5.43	0.00	100.00	142,949	CPS ASEC
Hourly wage	8.56	7.16	0.00	99.99	22,380	PSID

Notes: The sample is restricted to observations in PDA-treated and PDA-control states between 1973 and 1984, from the CPS ASEC (panel a) and PSID (panel b). All statistics are computed using individual sample weights. Wage income is measured in US dollars and is pre-tax. Hourly wage income is constructed using annual pre-tax wage income, number of weeks employed and usual weekly working hours. Pregnant is defined as reporting, in March of the following year, having a child younger than one with the CPS ASEC and as having given birth in the current year with the PSID. *The PSID records as a single category "Quit/resigned/retired/pregnancy/job change".

Table 2: Difference-in-Differences Estimates, CPS ASEC data

	Employed (1)	Hired (2)	Hourly Wage (3)	Pregnant (4)	Pregnant and Fired (5)
PDA-treated x Post-PDA	-0.045	-0.001	-0.020	0.000	0.004
	(0.007)	(0.007)	(0.018)	(0.004)	(0.004)
Observations	115,217	34,395	68,374	115,217	10,441
Pre-PDA Mean	0.686	0.0873	3.549	0.0901	0.00440
R-squared	0.073	0.016	0.157	0.045	0.007
State FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓

Notes: The sample is restricted to female respondents age 18 to 35 in PDA-treated and PDA-control states between 1973 and 1984, from the CPS ASEC. Estimates are computed using ASEC individual sample weights and include the fixed effects indicated in the table. Standard errors, in parentheses, are robust to heteroskedasticity. Employed is a dummy equal to 1 if the respondent is employed at some point in the year. Hired is a dummy equal to 1 if the respondent was not employed in the previous year and is employed in the previous week, and 0 if not employed in the previous week. Wage income is measured in US dollars and is pre-tax. Hourly wage income is constructed using annual pre-tax wage income, number of weeks employed and usual weekly working hours, and expressed as a z-score. Pregnant is defined as reporting, in March of the following year, having a child younger than one. Fired is a dummy equal to 1 if an individual absent from work in the previous week is absent due to a layoff and 0 otherwise.

Table 3: Difference-in-Differences Estimates, PSID data

	Employed	Hired	Hourly Wage	Pregnant
	(1)	(2)	(3)	(4)
PDA-treated x Post-PDA	-0.069	-0.053	0.008	-0.006
	(0.022)	(0.037)	(0.060)	(0.013)
Observations	19,197	7,060	12,080	17,786
Pre-PDA Mean	0.660	0.279	3.845	0.0948
R-squared	0.087	0.048	0.237	0.044
State FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓

Notes: The sample is restricted to female respondents age 18 to 35 in PDA-treated and PDA-control states between 1973 and 1984, from the PSID. Estimates are computed using individual sample weights and include the fixed effects indicated in the table. Standard errors are robust to heteroskedasticity. Employed is a dummy equal to 1 if the respondent is employed at some point in the year. Hired is a dummy equal to 1 if the respondent was not employed in the previous year and is employed in the current year, and 0 if not employed in the current year. Hourly wage income is the average pre-tax wage earnings per hour, expressed as a z-score. Pregnant is defined as a dummy for having a child born in the current year.

Table 4: Changes in Hourly Wage and Employment, Heterogeneity by Share Female in Industry, CPS ASEC data

	Hourly Wage in Industry			Employed in Industry		
	All	$\geq 75\%$	$< 75\%$	All	$\geq 75\%$	$< 75\%$
	(1)	(2)	(3)	(4)	(5)	(6)
PDA-treated x Post-PDA	-0.020	-0.062	0.001	-0.045	-0.008	-0.037
	(0.018)	(0.030)	(0.023)	(0.007)	(0.007)	(0.008)
Observations	68,374	22,442	45,921	115,217	115,217	115,217
Pre-PDA Mean	3.549	3.191	3.725	0.686	0.221	0.465
R-squared	0.157	0.157	0.153	0.073	0.013	0.048
State FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓

Notes: The sample is restricted to women of age 18 to 35 in PDA-treated and PDA-control states between 1973 and 1984, from the CPS ASEC. Columns (1) to (3) further restrict the sample to employed individuals and estimate our baseline specification (7) using hourly wage as dependent variable. In columns (2) and (3), the specification is estimated using only respondents employed in industries with a share of women employed included in the interval reported at the top of the table. Column (4) uses a dummy for employment as outcome. Columns (5) and (6) use a dummy equal to one if the respondent is employed in an industry with a share of women employed included in the interval reported at the top of the table, and zero otherwise. Estimates are computed using ASEC individual sample weights and include the fixed effects indicated in the table. Standard errors are robust to heteroskedasticity. The hourly wage is expressed as a z-score. The hourly wage is constructed using annual pre-tax wage income, number of weeks employed and usual weekly working hours.

APPENDIX

A.1 Data on Pregnancy-Discrimination Policies

To identify the effects of the PDA, we must determine which states passed comparable or more protective policies before 1978. To do this, we carefully studied the provisions of the PDA and their interpretation in the Law literature[42]. Moreover, we collected and codified information on all pre-1978 state policies regarding the employment of pregnant women, which we discussed in section 3. We compiled these data using both primary and secondary sources, guided by policy reports from the National Partnership for Women and Families, the US Census Bureau, and the Department of Labor (see here). We obtained the original text for almost all of these state laws, administrative rulings and court decisions. This was mostly done by consulting legal databases. In particular, we used the HeinOnline database on Session Laws, which provided us with specific information on the date of enactment, exact provisions and amendments of these policies. For administrative rulings, this was done by contacting the relevant state archives and libraries. For the few cases in which it was not possible to recover the original text, we collected information on their provisions and coverage from journal articles about them. In particular, we obtained these data from Stucke (1945), Dowd (1985), Gardin and Richwald (1986), and O'Brien and Madek (1988). Together, these data allowed us to reconstruct the steps that preceded the passage of the PDA of 1978, described above in section 3, and to build a viable reference group for states first 'treated' in 1978, as detailed in section 5. The complete list of these states and their state policies can be found in Table A2.

A.2 Conceptual Framework: Derivations and Extensions

A.2.1 Model Derivations

Equilibrium. For a given C, an interior competitive equilibrium is an allocation $(\mathbf{n}, \mathbf{w})$ such that the firm maximizes (1) at $\mathbf{w}$ and each labor market clears: $w(g,z) = W(g,z,n(g,z))$. Concavity makes the firm's first-order conditions sufficient. Thus an interior equilibrium solves

$$MP(g,z;\mathbf{n}) = W(g,z,n(g,z)) + q(g,z)C, \qquad \forall\,(g,z). \tag{A1}$$

We assume existence of an interior equilibrium at the levels of fertility costs considered. The following paragraph proves that such an interior equilibrium is unique.

Suppose distinct interior allocations $\mathbf{n}$ and $\hat{\mathbf{n}}$ solve (A1) at the same C. Subtracting their

[42] See for instance Weissmann (1983), Siegel (1985), Remmers (1989), and Habig (2007).

equilibrium conditions and taking the inner product with their employment difference yields

$$\begin{aligned} 0 &< \sum_{g,z} [n(g,z) - \hat{n}(g,z)][W(g,z,n(g,z)) - W(g,z,\hat{n}(g,z))] \\ &= \sum_{g,z} [n(g,z) - \hat{n}(g,z)][MP(g,z;\mathbf{n}) - MP(g,z;\hat{\mathbf{n}})] \leq 0 \end{aligned}$$

Strictly increasing inverse supply gives the first inequality, and concavity of production gives the last one. The contradiction establishes uniqueness of interior employment and, consequently, wages.

Equilibrium response to an increase in C. We now determine equilibrium responses to a marginal increase in C. We do this by totally differentiating the equilibrium system (A1) with respect to $\mathbf{n}$ and C, around the baseline equilibrium. We maintain the assumption that all other primitives are unchanged by the increase in C.

For each pair (g,z), we obtain

$$\sum_{h,s} \frac{\partial MP(g,z;\mathbf{n})}{\partial n(h,s)}\, dn(h,s) - \frac{\partial W(g,z,n(g,z))}{\partial n(g,z)}\, dn(g,z) - q(g,z)\, dC = 0. \tag{A2}$$

Let $\mathbf{q} = \{q(g,z)\}_{(g,z)}$ and define

$$\mathbf{J} \equiv \mathrm{diag}_{g,z}\{\frac{\partial W(g,z,n(g,z))}{\partial n(g,z)}\} - \nabla^2_{\mathbf{n}} Y(\mathbf{n}), \tag{A3}$$

where $\nabla^2_{\mathbf{n}} Y(\mathbf{n})$ is the Hessian of the production function Y, whose entries are the first derivatives of the marginal products in (A2). We can stack the conditions (A2) and write the system as

$$\mathbf{J}\, d\mathbf{n} = -\mathbf{q}\, dC, \tag{A4}$$

The first term in $\mathbf{J}$ is positive definite since the labor supply has strictly positive slope. Concavity of Y implies that its Hessian is negative semi-definite. Hence $\mathbf{J}$ is symmetric positive definite and invertible.

Since Y is twice continuously differentiable and each inverse labor-supply function $W(g,z,\cdot)$ is continuously differentiable, the equilibrium conditions are continuously differentiable in $(\mathbf{n}, C)$.

We can thus use the implicit-function theorem to express the equilibrium employment allocation $\mathbf{n}$ as a locally unique, continuously differentiable function of C, around the baseline

equilibrium. Its derivative is

$$\frac{d\mathbf{n}}{dC} = -\mathbf{J}^{-1}\mathbf{q}. \tag{A5}$$

Define the diagonal entry of $\mathbf{J}$ for a given pair (g, z) as

$$\mathcal{J}(g,z) \equiv \frac{\partial W(g,z,n(g,z))}{\partial n(g,z)} - \frac{\partial MP(g,z;\mathbf{n})}{\partial n(g,z)} > 0,$$

which is positive because $\mathbf{J}$ is positive definite. We thus obtain the employment response for workers with characteristics (g, z), as written in (4)

$$\frac{dn(g,z)}{dC} = \frac{-q(g,z) + \sum_{(h,s)\neq(g,z)} \frac{\partial MP(g,z;\mathbf{n})}{\partial n(h,s)} \frac{dn(h,s)}{dC}}{\mathcal{J}(g,z)}. \tag{A6}$$

The first term in the numerator captures direct exposure to the increased fertility costs, which in this stylized framework affects only fertile-age women. The second term captures changes in own marginal product induced by employment adjustments in other worker types. The denominator scales these effects and is a function of the slope of the inverse labor supply and of the the derivative of marginal product with respect to the type's own employment.

Decline in risk-weighted employment. We now show the derivations behind the result discussed in section 2, that pregnancy-risk-weighted employment necessarily declines as C increases and that this implies the negative response of fertile-age women's employment and wages.

Pre-multiplying the equilibrium response system (A5) by $\mathbf{q}'$ gives

$$-\mathbf{q}'\mathbf{J}^{-1}\mathbf{q} = \sum_{g,z} q(g,z)\frac{dn(g,z)}{dC} < 0. \tag{A7}$$

The strict inequality follows from $\mathbf{J}^{-1}$ being positive definite and $\mathbf{q} \neq \mathbf{0}$. That is, the employer's assessed probability of pregnancy is positive for at least one worker type. Thus pregnancy-risk-weighted employment strictly declines.

In this model, only one worker type has positive $q(g, z)$, fertile-age women. Thus, A7 simplifies to

$$q(f,0)\frac{dn(f,0)}{dC} < 0, \qquad \text{and therefore} \qquad \frac{dn(f,0)}{dC} < 0.$$

Strictly increasing inverse labor supply then implies $dw(f, 0)/dC < 0$.

A.2.2 Gender Gap in Employment-to-Population Ratio

Let $P(g,0) > 0$ be fixed fertile-age populations. The empirical gap is

$$\Gamma^{EP}(C) = \frac{n(m,0;C)}{P(m,0)} - \frac{n(f,0;C)}{P(f,0)}. \tag{A8}$$

Define $\mathcal{S}^{EP}$ by replacing Γ with Γ^{EP} in (5). The identity and lower-bound equivalence remain valid as long as the post-shock gap, expressed in employment-to-population, remains positive. That is, male employment-to-population rates are higher than female ones, which is verified in our empirical application. Equal fixed populations simply rescale the headcount gap and do not affect shares. In the empirical implementation, we use the gap in employment-to-population ratios, so we do not need to assume fixed equal populations.

A.2.3 PDA Cost Mapping and Retention Decision

Timing, beliefs, and costs. This dynamic extension provides a foundation for the expression of fertility costs $C(\phi) = \min\{\phi, R\}$, presented in section 3.1. An active worker becomes pregnant next period with probability $q(g,z)$.[43]

Employers discount time by $\beta \in (0,1)$. Let $d \geq 0$ be the dismissal cost evaluated at the time of pregnancy (i.e., not in present-value). A retained pregnant worker does not contribute to active production and generates a net flow cost $\kappa \geq 0$. We assume that she returns to the nonpregnant state next period with probability $\delta \in (0,1]$ and otherwise remains in the pregnant state. Firms can reconsider dismissal in each pregnancy period.[44]

Consistent with the main text, ϕ and R are discounted to the current active employment decision, one period before pregnancy:

$$\phi \equiv \beta d, \qquad R \equiv \frac{\beta\kappa}{1-\beta(1-\delta)}. \tag{A9}$$

We assume that the PDA raises only d, and hence ϕ.

Retention decisions in equilibrium. Let V^A denote the value of a marginal active relationship, and V^P its value at pregnancy before the dismissal decision. We omit type indices for brevity. Active relationships can be freely created and terminated at market wages, without other adjustment costs. Firms can freely create and terminate active employment

[43] We assume employers hold accurate fertility expectations. Thus $q(g,z)$ is both assessed risk and the actual conditional transition probability. This simplification avoids introducing a separate probability, without affecting the predictions of the model.

[44] We assume that there are no frictions to hiring and firing, except pregnancy protections, so dismissal of non-pregnant workers is always possible at no cost.

relationships, so in an interior stationary equilibrium it must be that $V^A = 0$. The pregnancy-state value solves

$$V^P = \max\{-d, -\kappa + \beta[\delta V^A + (1-\delta)V^P]\}. \tag{A10}$$

With $V^A = 0$, retention of pregnant workers costs firms $\kappa/[1-\beta(1-\delta)]$, conditional on pregnancy. Dismissal is optimal below this threshold, and retention above it. Therefore, in equilibrium, we have that

$$V^P = -\min\left\{d, \frac{\kappa}{1-\beta(1-\delta)}\right\}. \tag{A11}$$

Since the active marginal relationship satisfies

$$V^A = MP(g,z;\mathbf{n}) - w(g,z) + \beta[(1-q(g,z))V^A + q(g,z)V^P]. \tag{A12}$$

we obtain the expressions of Section 3.1

$$C = -\beta V^P = \min\{\phi, R\}, \qquad MP(g,z;\mathbf{n}) = w(g,z) + q(g,z)C. \tag{A13}$$

The PDA-induced variation in fertility costs. For initially dismissing firms, $\phi_0 < R$. Raising dismissal costs to $\phi_1 > \phi_0$ changes optimized fertility costs by

$$\Delta C = \min\{\phi_1, R\} - \phi_0 = \begin{cases} \phi_1 - \phi_0, & \phi_1 \leq R, \\ R - \phi_0, & \phi_1 > R. \end{cases} \tag{A14}$$

Thus $0 < \Delta C \leq R - \phi_0$.

In the baseline model, we assume that firms continue dismissing workers upon pregnancy. This is consistent with the historical evidence described in Section 3 and the descriptive statistics discussed in Section 5. It also simplifies the exposition because it ensures that currently employed and actively employed, non-pregnant, workers coincide. Allowing firms to switch to retention does not change the model unambiguous predictions for employment and wages of non-pregnant fertile-age women, the group that is directly affected by the increase in expected-fertility discrimination. However, it requires keeping track separately of actively and inactively employed workers and make sure they are consistent with labor supply. For this reason, we next extend the framework to account for pregnant and non-pregnant labor supplies.

Population stocks. Let $P(g,z)$ be the population of type (g,z), assumed fixed and exogenous. We also assume that the probability of pregnancy at the beginning of the next

period is $q(g,z)$ is the same for employed and non-employed workers, as long as they are not currently pregnant.[45] And that pregnant non-employed workers exit the pregnancy state with the same probability δ as retained workers. That is, flows into and out of the pregnancy state do not depend on the worker's employment status.

Let $M(g,z)$ be the nonpregnant population available for active work, and $H(g,z)$ the pregnant inactive population, including both employed and non-employed workers. For types with no pregnancy risk, we assume $M = P$ and $H = 0$. In a stationary equilibrium they are related by

$$H(g,z) = \frac{q(g,z)}{\delta} M(g,z), \qquad M(g,z) = \frac{\delta}{\delta + q(g,z)} P(g,z). \tag{A15}$$

So that pregnancy inflows qM are equal to exits δH.

Retained pregnant employment. Let $p(g,z)$ denote the stock of retained pregnant employees and $e(g,z)$ the total employment stock. In the baseline model we assume that in equilibrium $\mathbf{p} = 0$ adn $\mathbf{e} = \mathbf{n}$, both before and after the PDA. When firms choose to retain pregnant workers, the stock evolves according to

$$p_{t+1}(g,z) = q(g,z) n_t(g,z) + (1-\delta) p_t(g,z). \tag{A16}$$

Every pregnancy among active workers enters the retained stock, and each retained pregnancy exits with probability δ. In stationarity then

$$p(g,z) = \frac{q(g,z)}{\delta} n(g,z), \qquad e(g,z) = n(g,z) + p(g,z). \tag{A17}$$

On returning to the non-pregnant state, workers reenter the pool for active employment, whose level can be freely adjusted by firms, since there are no hiring or firing frictions for non-pregnant workers.

Consider now the case of a switch from strict dismissal before the PDA to strict retention after it. Denote stationary active employment in the pre- and post-PDA period by n_0 and n_1. For fertile-age women,

$$e_1 - e_0 = n_1 - n_0 + \frac{q(f,0)}{\delta} n_1 > n_1 - n_0. \tag{A18}$$

Thus the observed headcount change exceeds the active-employment change by a positive amount. Using estimates based on total headcount employment, we would thus underestimate

[45] The overall average probability of pregnancy among employed and non-employed is allowed to differ and will depend on the differential selection into employment across (g,z).

the decline in active employment due to the increase in expected-fertility discrimination. The increase in retention is an important response and a margin that the PDA was designed to directly address. However, it is separate from changes in expected-fertility discrimination which, by definition, involve differential treatment based on expectations of future fertility, rather than current realized fertility.

Retaining pregnant workers will not change the response of other worker types, for a given increase in fertility costs. Define the observed gap in total headcount employed relative to population as

$$\Gamma^{EP,e} = e(m,0)/P(m,0) - e(f,0)/P(f,0).$$

Its change satisfies

$$\Delta\Gamma^{EP,e} = \Delta\Gamma^{EP} - \frac{q(f,0)n_1}{\delta P(f,0)}, \tag{A19}$$

where $\Delta\Gamma^{EP}$ only includes actively employed workers in the numerator. Retention therefore reduces the observed widening of the gap relative to the active-employment comparison.

A.2.4 PDA Identifying Restrictions and Alternative Channels

In Section 3.1, we framed the PDA as an increase in costs of fertility to firms, which can identify expected-fertility discrimination since it only affects labor market outcomes through this channel. In this section, we consider plausible violations of this exclusion restriction and argue that they are unlikely to undermine the validity of our results.

Fertility expectations. We first consider the possibility that the PDA affected employer expectations about future fertility. While the estimates presented in section 6 show no effects of the PDA on fertility rates, it is possible that employers expected the PDA to lower the opportunity costs of fertility and thus raise the probability of future pregnancies. An increase in $q(g,z)$ would violate the exclusion restriction of a pure shock to fertility costs C. However, the effect of the PDA would still operate through expected-fertility discrimination. To see this, notice that the expected-fertility wedge would increase by

$$q_1(g,z)C_1 - q_0(g,z)C_0 = q_0(g,z)\Delta C + C_0\Delta q(g,z) + \Delta q(g,z)\Delta C. \tag{A20}$$

Hence, the model predictions would remain qualitatively unchanged and our interpretation of the decline in employment and wages of fertile-age women as proof of increased expected-fertility discrimination would still hold.

Labor supply. We next consider the case in which fertile-age women value employment protection provided by the PDA, as a non-wage amenity, and thus see an expansion in their labor supply. Notice that, even if dismissal upon pregnancy remained frequent, women could still value the possibility of being financially compensated, thanks to the newly introduced sanctions. Let τ index exposure to the PDA and allow inverse labor supply for fertile-age women to depend directly on it, $W(f, 0, n; \tau)$. Define

$$b \equiv -\frac{\partial W(f, 0, n; \tau)}{\partial \tau} \geq 0, \qquad C_\tau \equiv \frac{dC}{d\tau} > 0,$$

where the partial derivative is evaluated at fixed employment. Thus b measures the outward supply shift as a reduction in the wage required to supply a given level of employment.

Differentiating the equilibrium conditions and allowing all labor inputs to adjust gives

$$\frac{dn(f,0)}{d\tau} = -\chi(f,0)[q(f,0)C_\tau - b], \qquad \chi(f,0) \equiv [\mathbf{J}^{-1}]_{(f,0),(f,0)} > 0. \tag{A21}$$

The coefficient $\chi(f, 0)$ incorporates adjustments in other labor markets and is positive because $\mathbf{J}^{-1}$ is positive definite. The cost increase reduces demand for fertile-age women, while the outward supply shift offsets part of this contraction.

Absent a supply shift, the employment response would be

$$\left.\frac{dn(f,0)}{d\tau}\right|_{\text{cost}} = -\chi(f,0)q(f,0)C_\tau < 0.$$

Consequently,

$$\frac{dn(f,0)}{d\tau} = \left(1 - \frac{b}{q(f,0)C_\tau}\right) \left.\frac{dn(f,0)}{d\tau}\right|_{\text{cost}}.$$

For $0 < b < q(f, 0)C_\tau$, employment still declines, but by less than with no outward shift in supply of fertile-age women.

Thus, while we have no evidence of increased labor supply in response to the PDA, at least in the period we examine, if an outward shift nevertheless occurred, an observed employment decline would understate the magnitude of the decrease due to the increase in fertility costs alone.

Maternity benefits and retention costs. As we discussed in Section 3 and examined empirically in Figure A10, the PDA affected employment of fertile-age women primarily by increasing employment protection and thus exposure to pre-existing employer-borne costs of retaining pregnant workers.

In settings where the PDA further expanded health insurance and temporary disability

leave benefits, it also increased costs conditional on retention. Then the optimized mapping is $C = \min\{\phi, R\}$ with both arguments potentially changing, and the fixed-R bound in (A14) cannot be applied using the old R. The presence of this additional channel can change the exact interpretation of the PDA as a cost shock. However, it does not violate the exclusion restrictions as its effects would still operate through an increase in fertility costs.

At the same time, as documented by Gruber (1994) and Timpe (2024), fertile-age women could value these benefits when provided. In this case, their labor supply would shift outward and the same argument of the previous paragraph would apply.

Retention of pregnant workers and observed employment. If the PDA induced firms to retain pregnant workers, the resulting increase in retained employment would partly offset the decline in active employment caused by increased expected-fertility discrimination, as shown in (A18). The retention *versus* dismissal decision is a response to realized fertility, different from the anticipatory employment decisions based on expected fertility, through which expected-fertility discrimination operates. Hence, estimates based on total employed headcount would understate the employment consequences of this increase in discrimination, including its contribution to widening the gender employment gap.

A.2.5 CES Technology and Isoelastic Supply for Calibration

Outer CES technology. For the quantification exercise, where we need to calibrate the model, we follow Card and Lemieux (2001) and impose a nested CES structure. Thus, we retain the inner aggregates (2) and restrict $\tilde{Y}$ to the case of constant-returns CES technology

$$\tilde{Y}(L(f), L(m)) = \left[\theta_f L(f)^\rho + \theta_m L(m)^\rho\right]^{1/\rho}, \quad \theta_g > 0, \quad \theta_f + \theta_m = 1, \quad \rho < 1, \quad \rho \neq 0. \tag{A22}$$

The elasticity of substitution between female and male aggregate effective labor is $\sigma_{fm} = 1/(1-\rho)$. For positive inputs,

$$r_g = \theta_g L(g)^{\rho-1} \tilde{Y}^{1-\rho}, \qquad Y_{fm} = (1-\rho)\theta_f \theta_m L(f)^{\rho-1} L(m)^{\rho-1} \tilde{Y}^{1-2\rho} > 0. \tag{A23}$$

So female and male effective labor are necessarily Edgeworth complements. This implies that

$$\Delta_{\text{GE}} = 1 - \Lambda_f Y_{ff} - \Lambda_m Y_{mm}, \qquad \frac{dr_f}{dC} = -\frac{\Omega_f Y_{ff}}{\Delta_{\text{GE}}}.$$

Both male age groups therefore contract in response to an increase in C, under this nested CES structure with constant returns to scale.

Isoelastic active labor supply. For each cell, specify

$$W(g,z,n) = \omega(g,z)\left(\frac{n}{M(g,z)}\right)^{1/\varepsilon(g,z)}, \quad \omega(g,z) > 0, \quad \varepsilon(g,z) > 0, \quad 0 < n < M(g,z), \tag{A24}$$

with slope

$$\frac{\partial W(g,z,n)}{\partial n(g,z)} = \frac{W(g,z,n)}{\varepsilon(g,z)n}. \tag{A25}$$

We denote by $M(g,z)$ the non-pregnant population available for active work. In the case of a switch to retention of pregnant workers, this population is given by (A15). The parameter $\omega(g,z)$ is the upper reservation-wage scale, and $\varepsilon(g,z)$ is the elasticity of active employment with respect to the wage below this cap. Equivalently, the reservation wage of non-pregnant workers is distributed according to $F_{g,z}(w) = (w/\omega(g,z))^{\varepsilon(g,z)}$ for $0 \leq w \leq \omega(g,z)$, and $F_{g,z}(w) = 1$ above that threshold.

A.3 Calibration and the Monotonicity Condition

In Section 7, we interpret the effect of the PDA on the gender employment gap as a lower bound on the total contribution of expected-fertility discrimination to this gap. This interpretation requires

$$\Gamma(C^{\text{NoPDA}}) \geq \Gamma(0).$$

A sufficient condition for this is that the gender employment gap among fertile-age individuals, $\Gamma(C)$, be weakly increasing in fertility costs C between 0 and C^{NoPDA}. Since the conceptual framework in Section 2 does not guarantee this condition analytically, we therefore assess the condition quantitatively by calibrating the model. To do this, we follow Card and Lemieux (2001) and impose a CES form also for the output function Y. For labor supplies, we assume an isoelastic specification. We formalize these restrictions in Appendix A.2.5.

In the firing regime considered in the calibration, C^{NoPDA} coincides with the effective dismissal costs ϕ absent the PDA. Define the expected-fertility wedge in this case for young women as

$$x \equiv q_f(0)C^{\text{NoPDA}}.$$

Since $q_f(0) > 0$, monotonicity of the gender employment gap in x is equivalent to monotonicity in ϕ. We denote by

$$x_0 = q_f(0)C^{\text{NoPDA}} \quad \text{and} \quad x_1 = q_f(0)C^{\text{PDA}}$$

the pre- and post-PDA wedges, respectively. The calibration therefore asks whether the equilibrium gender employment gap is increasing over $x \in [0, x_0]$. In fact, we evaluate the

stronger condition over the full interval $[0, x_1]$. Since neither x_0, nor x_1 are identified, we assess this condition over a range of possible values that are consistent with the observed effects of the PDA and values of the other primitives that have been established in the literature. We explain this in detail below.

Calibration. The calibration is set-valued: rather than selecting a single parameter vector, we consider a range of parameterizations disciplined by the literature and retain those that reproduce the estimated employment responses to the PDA. Table A1 reports the six structural parameters disciplined externally, together with their benchmark values and the ranges considered. The aggregate male and female extensive-margin labor-supply elasticities are based on Elder et al. (2025); the age profile of female labor supply is motivated by Attanasio et al. (2018); and the substitution elasticities in the within-gender and across-gender CES aggregators are based on Card and Lemieux (2001) and Acemoglu, Autor, et al. (2004), respectively. Because the literature provides less guidance on the relative labor-supply elasticity of younger and older men, we use equality as the benchmark and consider a deliberately wide range around it.

Table A1: Externally disciplined calibration parameters

Parameter	Description	Benchmark	Range	Source
$\bar{\varepsilon}_f$	Avg. female extensive-margin labor-supply elasticity	1.48	$[1.30, 1.65]$	Elder et al. (2025)
$\bar{\varepsilon}_m$	Avg. male extensive-margin labor-supply elasticity	0.085	$[0.04, 0.13]$	Elder et al. (2025)
R^{ε}_f	Young/old female elasticity ratio	1.30	$[1.00, 1.50]$	Attanasio et al. (2018)
R^{ε}_m	Young/old male elasticity ratio	1.00	$[0.50, 2.00]$	Benchmark restriction
σ_{age}	Within-gender age substitution	5.0	$[4.0, 6.0]$	Card and Lemieux (2001)
σ_{fm}	Male/female substitution	3.5	$[2.4, 4.2]$	Acemoglu, Autor, et al. (2004)

The remaining calibration inputs are constructed from the CPS. We use individuals in PDA-treated states and define the pre-PDA period as 1975–1978 and the post-PDA period as 1981–1984.[46] Pre-PDA population shares, employment rates, employment headcounts, and wages anchor the baseline equilibrium at x_0. For each candidate vector of externally disciplined parameters and policy wedges (ψ, x_0, x_1), the model then determines the cell-specific labor-supply elasticities, supply intercepts, technology shares and scale, and equilibrium employment

[46]These do not coincide with the periods used in most of the main analyses for two reasons. First, hourly wages are only observed starting in 1975. Second, we omit the two years just after the PDA, 1979-1980, to better capture the new equilibrium.

and wages. In particular, post-PDA wages are equilibrium outcomes and are not independently targeted in the calibration.[47]

We select values of (x_0, x_1) using the estimated PDA effects on employment in the four gender-by-age cells. The estimated employment-rate effects are converted into fixed-population headcount effects using the corresponding pre-policy reference populations. Let $\hat{\Delta\mathbf{n}}$ denote the resulting vector of four estimated headcount effects and $\hat{\mathbf{V}}_n$ its covariance matrix. For each candidate calibration, we compute

$$Q = \left(\Delta\mathbf{n}^{\text{model}} - \hat{\Delta\mathbf{n}}\right)' \hat{\mathbf{V}}_n^{-1} \left(\Delta\mathbf{n}^{\text{model}} - \hat{\Delta\mathbf{n}}\right). \tag{A26}$$

Since $\hat{\mathbf{V}}_n$ has rank four, we retain parameter-policy tuples satisfying

$$Q \leq \chi^2_{4,0.95} = 9.49.$$

Conditional-wage treatment effects are not used in this acceptance criterion and provide an out-of-sample check on the wage responses generated by the model.

Monotonicity. The search considers 4,125 parameter-policy records. Of these, 416 satisfy the employment-based criterion in (A26), corresponding to 390 numerically distinct (ψ, x_0, x_1) tuples. Acceptance does not select a narrow region of the structural parameters disciplined by the literature. All 40 structural parameter draws considered admit at least one policy-wedge pair satisfying the empirical criterion, and the accepted tuples span the benchmark values and nearly the full robustness ranges reported in Table A1. The rejected records therefore primarily reflect combinations of structural parameters and policy wedges (x_0, x_1) that cannot jointly reproduce the four estimated employment responses, rather than rejection of particular values of the externally disciplined parameters. For each retained tuple, we solve the equilibrium along the full path from $x = 0$ to $x = x_1$ and compute the gender employment gap at ages 18 to 35[48]

$$\Gamma(x) = \frac{n_{m0}(x)}{P^{ref}_{m0}} - \frac{n_{f0}(x)}{P^{ref}_{f0}}.$$

The required monotonicity condition holds for every retained calibration. In particular,

$$\inf_{x \in [0, x_1]} \frac{d\Gamma(x)}{dx} > 0$$

[47] This is because wages observed after the PDA refer to a different selection of workers, as employment is directly affected by the PDA.

[48] Here 0 indexes $z = 0$, young workers, not the policy status x.

for all 416 accepted records, with the minimum derivative along a path ranging from 0.0753 to 0.1507 across the accepted set. Thus the gender employment gap is strictly increasing throughout the entire interval $[0, x_1]$ for every empirically admissible parameterization considered, and therefore also over the smaller interval $[0, x_0]$ relevant for the lower-bound argument.

A.4 Additional Details on the PSID

The availability of women's individual employment information in the PSID changes over our sample period. Through 1978, the PSID generally collected contemporaneous employment-status information only for the Reference Person (formerly the Head of Household), who is disproportionately male in our sample. An important exception is 1976, when the PSID conducted a separate interview with the spouse, who is almost always female in our sample ($\approx$99%). Beginning in 1979, employment information was collected much more systematically for other adult household members as well. Consequently, during the pre-PDA period, contemporaneous employment status reported directly by women is available primarily for women who are themselves the household Reference Person.

The PSID nevertheless provides additional information on wives before 1979 through the family interview. In particular, the Reference Person was asked retrospectively about employment and labor earnings during the previous calendar year both for himself and, when present, for his wife. We use these retrospective measures to supplement women's directly reported information when the latter is unavailable, allowing us to construct a substantially more complete time series of employment and earnings. This is the employment variable we use as outcome and on which we base the construction of the hiring outcome, defined as in the CPS ASEC. Hourly wage earnings, constructed as the average using total hours worked and total wage earnings in the year, are also reported retrospectively in the family interview.

Our findings are more precise under these comprehensive measures that use all the information available on female respondents' wages and employment outcomes. However, results are robust to using only employment information reported directly by women, restricting the sample to female Reference Persons only, or relying only on the measures obtained from the family interview.

Job separations. The longitudinal structure of the PSID also allows us to measure job separations more comprehensively than in the CPS ASEC. In particular, the PSID can identify and date separations even when an individual is employed in two adjacent survey years but changes employers between them. It also records information on the reason for the separation, allowing us to distinguish among different types of employment transitions.

Coverage of these separation measures is substantially more limited for women in the early part of our sample. Detailed separation information for women who are not the household Reference Person is available in 1976 and, more systematically, beginning in 1979. There are therefore too few pre-PDA observations with comparable separation information to obtain precise difference-in-differences estimates of post-pregnancy separations. We instead use these variables primarily to describe the incidence and composition of job separations following pregnancy or childbirth in the post-PDA period. These statistics are reported in table 1.

A.5 Triple Difference Estimator

As said in section 5, we augment the baseline DID specification of equation 7 relying on the fact that the policy is expected to affect mainly women in fertile ages who are pregnant or may be pregnant in the future. Instead, the policy should not directly affect older women and men, if not through substitution channels, that we deem much less strong. Indeed, as documented in section 6.3, we do not detect significant employment effects for these groups. We exploit this characteristic of pregnancy-discrimination policies to strengthen our empirical strategy adding a further difference: the one between outcomes of women aged 18 to 35 and of women 40 to 60 and men of ages 18 to 60. We thus do a triple-difference-in-differences strategy, in reverse. Let $I_{i,s,t}$ be an indicator taking value of one if the individual is a woman aged 18 to 35 and zero if she/he belongs to one of the other two groups. We augment specification (7) estimating

$$y_{i,s,t} = \beta T_s \times P_t \times I_{i,s,t} + \alpha T_s \times I_{i,s,t} + \nu P_t \times I_{i,s,t} + \rho I_{i,s,t} + \Gamma X_{i,s,t} + \eta_{s,t} + \varepsilon_{i,s,t} \qquad \text{(A27)}$$

where $\eta_{s,t}$ are state-by-year fixed effects and all other terms are as described above. Notice that the interaction between the PDA-treated and post-PDA indicators is subsumed by state-by-year fixed effects. If pregnancy-discrimination laws affect only women aged 18 to 35, and not the other included individuals, then β in (A27) identifies the parameter of interest under a less stringent parallel trends assumption. With the inclusion of a control group within each state, we can include in the specification state-by-year fixed effects and linearly control for all factors that vary across time and states and affect outcomes of fertile-age women and the control group in the same way. For instance, this would capture the effect of state-specific trends in labor market conditions. We thus only require that the within-state difference between outcomes of treated and control individuals would have evolved in parallel in PDA-treated and PDA-control states, had they both been treated before 1978. This is less stringent than the parallel trends presented above, but relies also on the assumption that the

control individuals are not affected by the enactment of the PDA.

A.6 Predicted Fertility

To confirm that the results are indeed driven by the PDA increasing the expected costs of employees' fertility, triggering a negative labor demand response, we show that the effects are much stronger for women who are more likely to become pregnant, based on their observable characteristics. We construct this measure using the relationship between observable characteristics and subsequent fertility in the PSID, and apply the estimated prediction functions to individuals in the CPS ASEC and Census.

Specifically, we use the PSID fertility histories to construct an indicator equal to one if an individual has at least one biological child in the following five years, from $t+1$ through $t+5$, and zero otherwise.[49] We estimate separate probit models for women and men using individual-year observations from all states and using pre-PDA years, 1970–1978, restricting the sample to individuals aged 18–60. Estimation uses the PSID individual fair-share cross-sectional weights.[50]

Both models include indicators for age in years; indicators for the number of biological children at the time of the survey (grouped into 0, 1, 2, and 3+); years since the most recent biological childbirth and this variable squared; an indicator for childbirth in the preceding year; marital status; educational attainment; and race. Years since the most recent childbirth is set to zero for individuals without children, whose status is separately captured by the indicators for the number of children.

For women, we assess predictive performance using five-fold cross-validation, assigning all observations of a given individual to the same fold. We estimate the model on four folds and predict childbirth probabilities for observations in the remaining fold, repeating this procedure across the five folds. Using these held-out predictions, we compare average predicted and realized fertility and calculate the Brier score and the area under the receiver operating characteristic curve, both for the full female estimation sample and for women aged 18–35. For women aged 18–35, we also assess calibration across predicted-risk deciles, estimate a calibration intercept and slope, and compare predictive performance with a model containing only age indicators.

The performance of this prediction exercise is generally good. As expected, age alone accounts for most of the variation across individuals and has strong predictive power. For

[49] We exclude individuals for which fertility in the following 5 years is not observed or available retrospectively from later years.

[50] The restriction to 1970–1978 applies to the years in which predictors are measured. Fertility outcomes can extend through 1983.

women, we assess predictive performance using five-fold cross-validation, keeping all observations of each individual in the same fold. Among women aged 18–35, the model achieves an out-of-fold area under the ROC curve of 0.758, compared with 0.700 for a model containing only age indicators; the weighted average predicted probability of childbirth within five years is 31.5%, close to the realized rate of 31.4%.

We use the models estimated on the full gender-specific PSID estimation samples to predict fertility for individuals observed in the CPS ASEC and in the Census. For an individual of sex g, the predicted probability is

$$\hat{p}_i = \Phi(Z_i'\hat{\alpha}_g)\,,$$

where Z_i contains the predictors described above and a constant, $\hat{\alpha}_g$ is the corresponding vector of estimated probit coefficients, and Φ is the standard normal cumulative distribution function.[51] We define "High Predicted Fertility", denoted by $HighPF_i$, as a predicted probability at or above the survey-weighted pre-PDA median in the relevant dataset and sample, calculated separately for women and men. These cutoffs are held fixed across the pre- and post-PDA periods.

The estimates displayed in Figure A7 are based on the following augmented version of equation 7:

$$\begin{aligned} y_{i,s,t} = {} & \beta_1 T_s \times P_t \times HighPF_i + \beta_2 T_s \times P_t \times (1 - HighPF_i) \\ & + \lambda_1 T_s \times HighPF_i + \lambda_2 P_t \times HighPF_i + \rho HighPF_i \\ & + \Gamma X_{i,s,t} + \theta_s + \delta_t + \varepsilon_{i,s,t}. \end{aligned} \tag{A28}$$

The coefficients β_1 and β_2 capture the difference-in-differences estimates for individuals with high and low predicted fertility, respectively. The additional interactions allow the high–low fertility gap to differ between treated and comparison states and to change after the PDA in both groups of states. The remaining variables are defined as in Equation 7.

The estimates reported in Table A11 are based on the following specification:

$$y_{i,s,t} = \beta P_t \times HighPF_i + \rho HighPF_i + \Gamma X_{i,s,t} + \theta_s + \delta_t + \varepsilon_{i,s,t}. \tag{A29}$$

This specification compares changes in outcomes between women age 18 to 35, with high and low predicted fertility, without using variation in PDA treatment status across states. The coefficient β measures the post-PDA change in the outcome gap between the two fertility

[51] The fertility-history variables are not measured identically across datasets. The PSID measures biological children, including children living outside the household, whereas the CPS ASEC and Census implementations use children residing in the household and their ages to approximate the corresponding predictors. Applying the PSID prediction function therefore requires comparability of these measures.

groups, conditional on the included controls and fixed effects. We estimate this specification separately for PDA-treated states, PDA-control states, and all states together.

A.7 Inference with a Small Number of Clusters

Inference in our difference-in-differences design is complicated by the small number of independent clusters at the level of treatment variation. Conventional cluster-robust inference relies on asymptotic approximations in the number of clusters and can perform poorly when the number of clusters is small (MacKinnon et al., 2023; Roth et al., 2023). This concern is particularly relevant when there is a substantial imbalance between the number of treated and comparison clusters. In our setting, the main sample contains 18 states, of which 15 are PDA-treated, leaving only three PDA-control states. Moreover, in the CPS ASEC, the available geographic identifiers group some states together, so that these states correspond to only 12 distinct state or state-group clusters, of which nine are treated.

There is no universal threshold at which conventional clustered inference becomes reliable, nor a single finite-cluster correction that is valid without additional assumptions. An often-cited rule of thumb from Angrist and Pischke (2009) is that approximately 42 clusters may be sufficient for conventional cluster-robust inference. Beyond this heuristic threshold, finite-sample performance also depends on features of the design such as cluster heterogeneity and the distribution of treatment across clusters (MacKinnon et al., 2023). Similarly, procedures for inference accounting for few clusters generally obtain improved inference by imposing additional restrictions on the distribution of cluster-level shocks or on the treatment assignment process (Roth et al., 2023). The plausibility of these restrictions is necessarily application-specific, but often hard to assess in practice.

For this reason, we report heteroskedasticity-robust standard errors in the main analyses as a baseline and use this appendix to document that our conclusions are robust to using state-level clustering and alternative procedures designed for a small number of clusters. We focus on our main effect of interest, employment of fertile-age women. Table A13 first reports, for reference, inference based on the heteroskedasticity-robust standard errors used in the rest of the paper. We then report conventional cluster-robust inference, clustering at the state level, or at the state-group level in the CPS ASEC. The conventional justification for these standard errors rely on the asymptotics in the number of clusters discussed above. With few clusters, the resulting variance estimator can be downward biased and can provide inaccurate inference.

We therefore implement two related finite-sample corrections to conventional cluster-robust inference. First, we use the bias-reduced adjustment proposed by Bell and McCaffrey

(2002), which modifies the cluster-robust variance estimator to reduce its finite-sample bias and adjusts the degrees of freedom and critical values. Second, we implement the extension of this approach proposed by Pustejovsky and Tipton (2018) to regressions with fixed effects. The resulting p-values are very similar across the two adjustments and are larger than those obtained using either heteroskedasticity-robust or conventional cluster-robust inference. In both cases, they remain below conventional significance levels.

A commonly used alternative for inference with few clusters is the wild cluster bootstrap. However, its validity with a fixed, small number of clusters requires additional restrictions. In particular, Canay et al. (2021) show that validity under few-cluster asymptotics relies on homogeneity-type restrictions on the distribution of the regressors and note that these conditions can be difficult to satisfy in difference-in-differences settings in which treatment varies at the cluster level. This concern is directly relevant to our main specification.[52]

As an alternative that does not rely on the conventional many-cluster approximation, we also implement the placebo inference procedure of Hagemann (2019). The procedure constructs cluster-specific estimates of the post-treatment change and compares the average estimate among treated clusters with the average among comparison clusters. The null distribution is then obtained by assigning placebo treatment status across clusters. This approach is designed for settings with a small number of large clusters and applies directly to difference-in-differences designs, accommodating as little as three clusters in a treatment group.

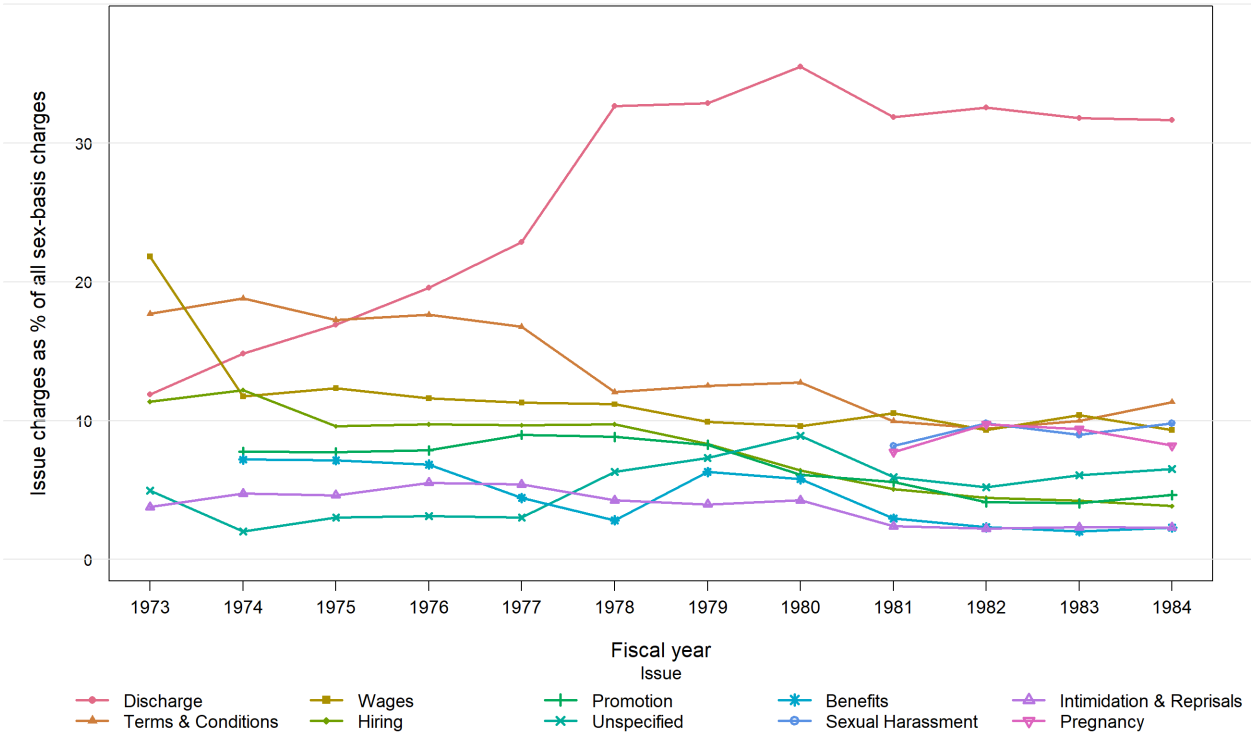
Because the placebo inference estimator is constructed from an equal-weighted comparison of cluster-specific estimates, its point estimate is not in general numerically equivalent to the coefficient from our pooled individual-level specification, which applies different weights across clusters due to different weighted samples. We therefore report the corresponding point estimate separately in Table A13. The qualitative pattern is stable across methods and data sources.

Thus, the alternative procedures designed to address few-cluster inference consistently support the negative employment pattern documented in the main analysis.

[52]See also Roth et al. (2023) for a discussion of the additional assumptions underlying bootstrap and randomization-based methods for DiD designs with few treated or comparison clusters.

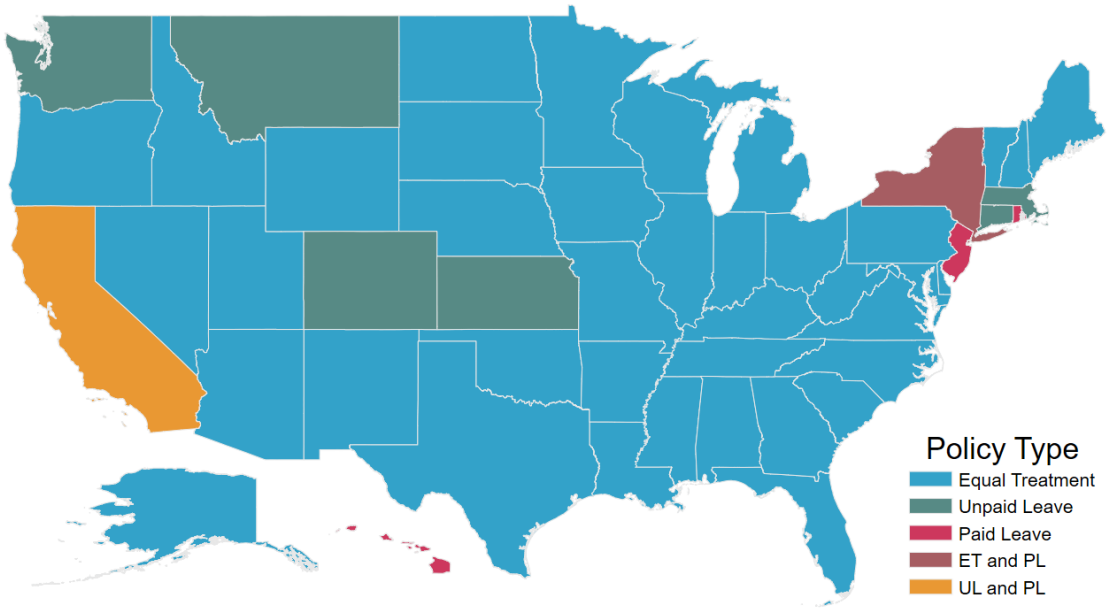
A.8 Additional Figures

Figure A1: Trends in EEOC Charges Against Private Employers



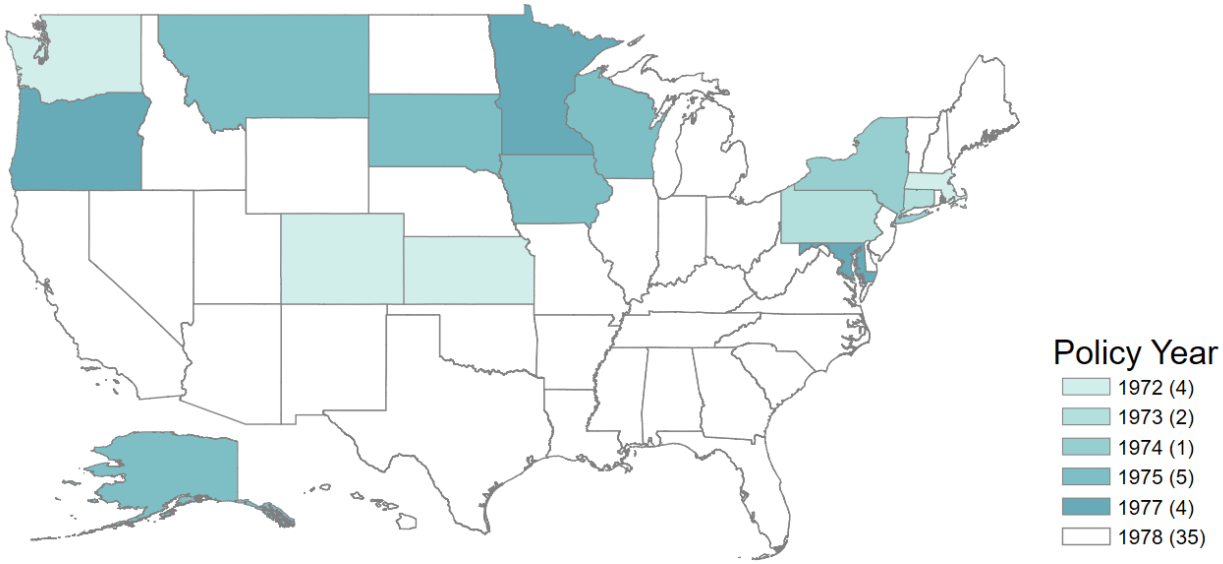
Notes: Own calculations based on the summary charge statistics reported in EEOC yearly reports, digitized by the authors. Each dot reports, for a given issue, the percentage of total sex-discrimination charges each year regarding that issue. Included are the 10 most frequent sex-discrimination issues in the 1973-1984 period. Each charge can involve multiple issues, issues are not mutually exclusive.

Figure A2: Type of Policy Adopted



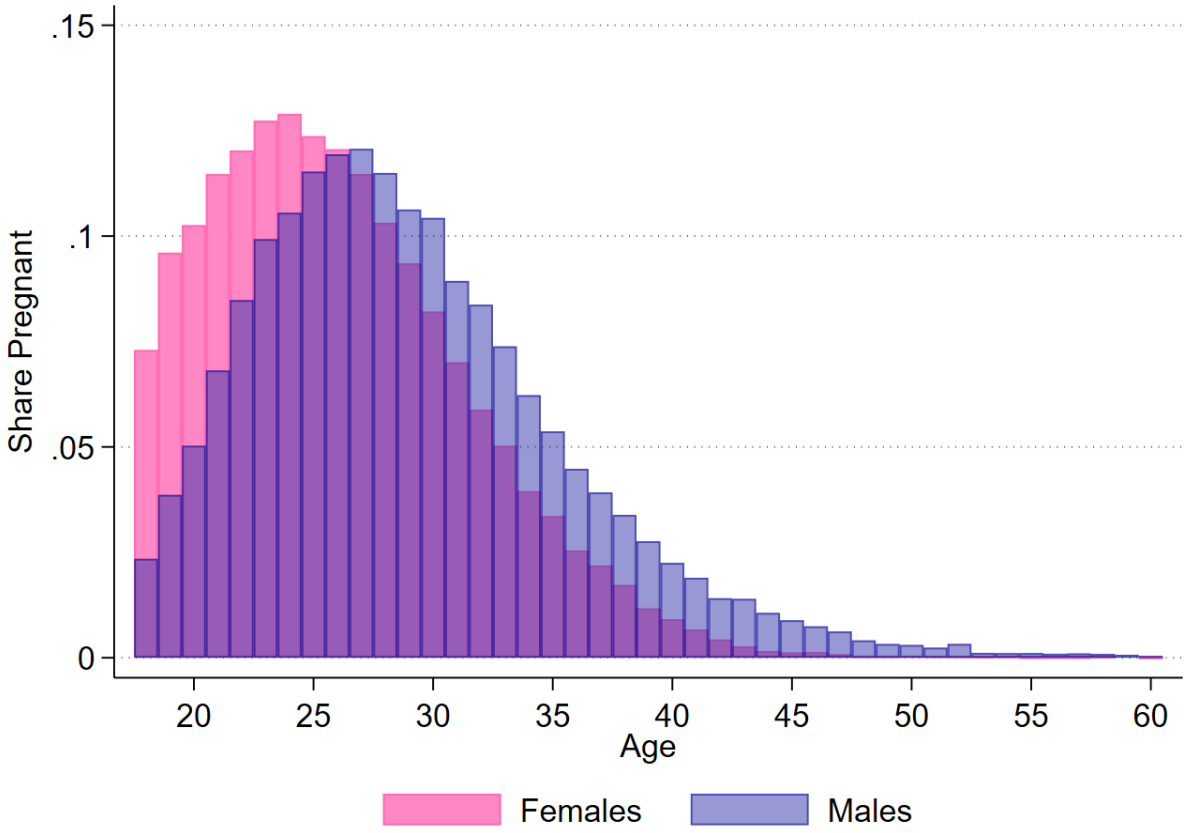
Notes: This map displays the type of policy adopted. States that adopted an equal treatment policy include also all states that adopted it only when the PDA was passed in 1978.

Figure A3: Timing of Pregnancy Discrimination Legislation



Notes: This map displays the year of passage of each state's pregnancy-discrimination policy. States that only enacted such a policy in 1978, with the passage of the PDA at the federal level are displayed in white. Michigan passed a state pregnancy-discrimination law in 1978, it is also displayed in white.

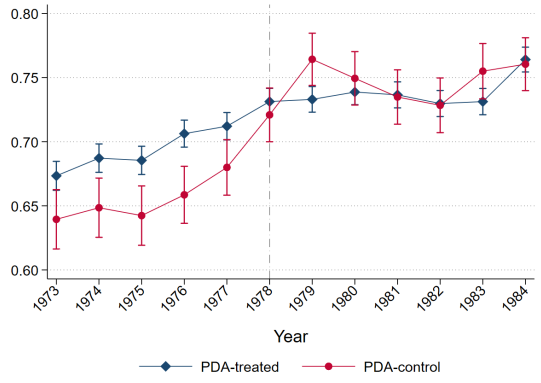
Figure A4: Share Pregnant by Age



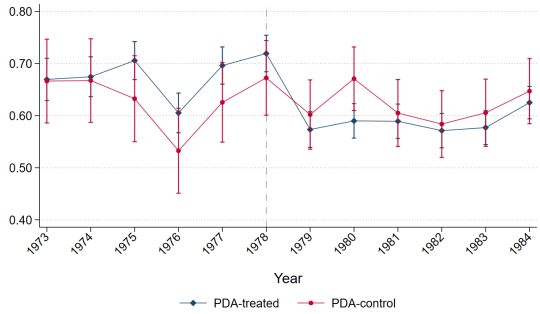
Notes: The sample consists of all CPS ASEC respondents of age 18 to 60, between 1973 and 1984. Estimates are computed using ASEC individual sample weights. Pregnancy for male and female respondents is defined as reporting having a child age 0 in March of the following year. For males the value is thus referred to the partner's pregnancy.

Figure A5: Raw Trends in Employment

(a) CPS ASEC



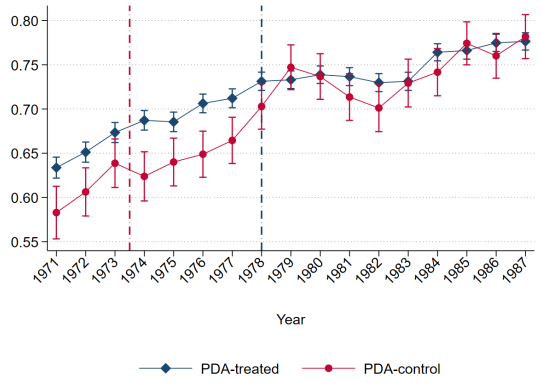
(b) PSID



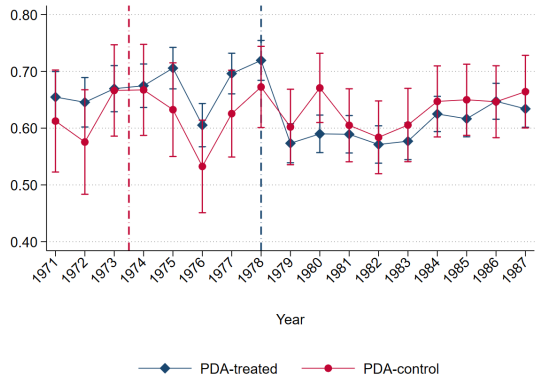
Notes: The sample consists of all female respondents of age 18 to 35 in PDA-treated and PDA-control states, between 1973 and 1984 from CPS ASEC, in panel (a), and PSID, in panel (b). Estimates are computed using each survey's individual sample weights. 95% confidence intervals are based on heteroskedasticity robust standard errors.

Figure A6: Raw Trends in Employment, Extended Time Window

(a) CPS ASEC

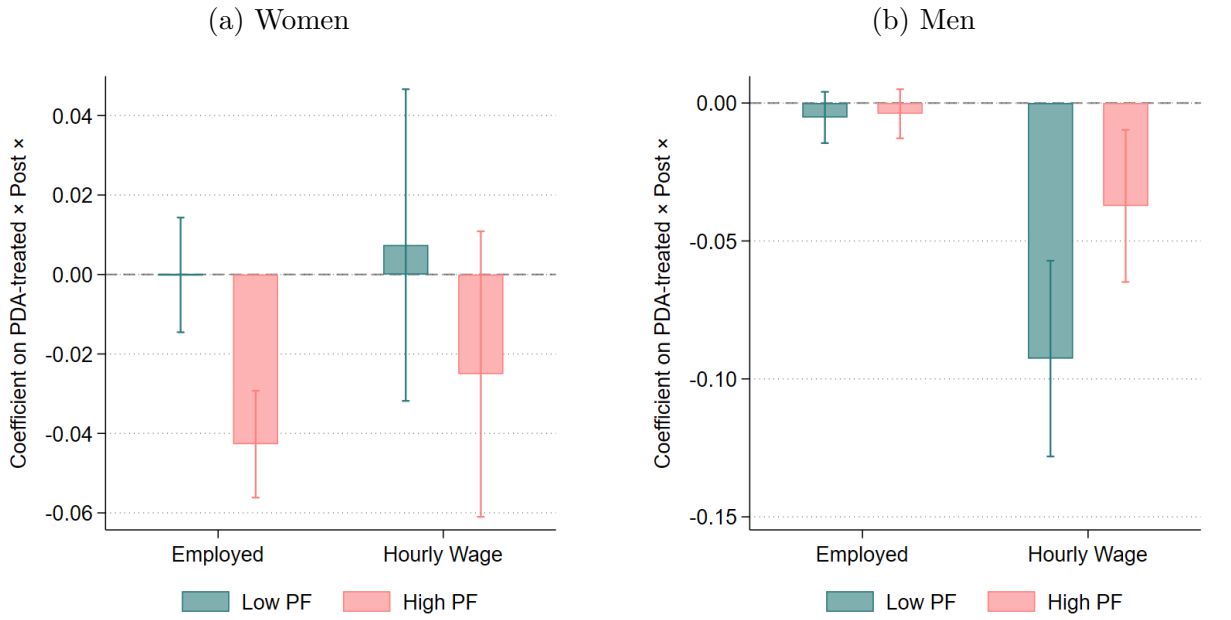


(b) PSID



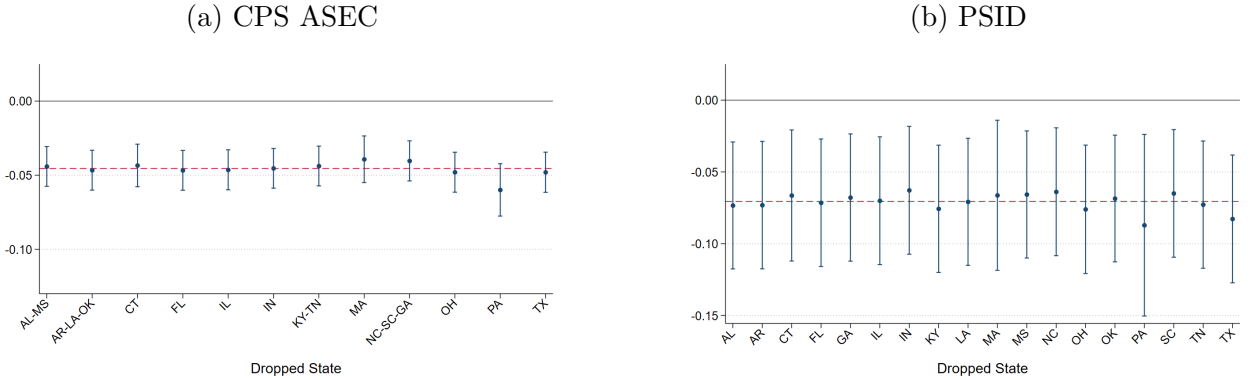
Notes: The sample consists of all female respondents of age 18 to 35, between 1971 and 1987 from CPS ASEC, in panel (a), and PSID, in panel (b). Estimates are computed using each survey's individual sample weights. The red (blue) vertical dashed line indicates the time at which PDA-control (PDA-treated) states get treated. PDA-control states include only Pennsylvania and Connecticut in panel (a), plus Massachusetts in panel (b).

Figure A7: Effects by Predicted Fertility, CPS ASEC



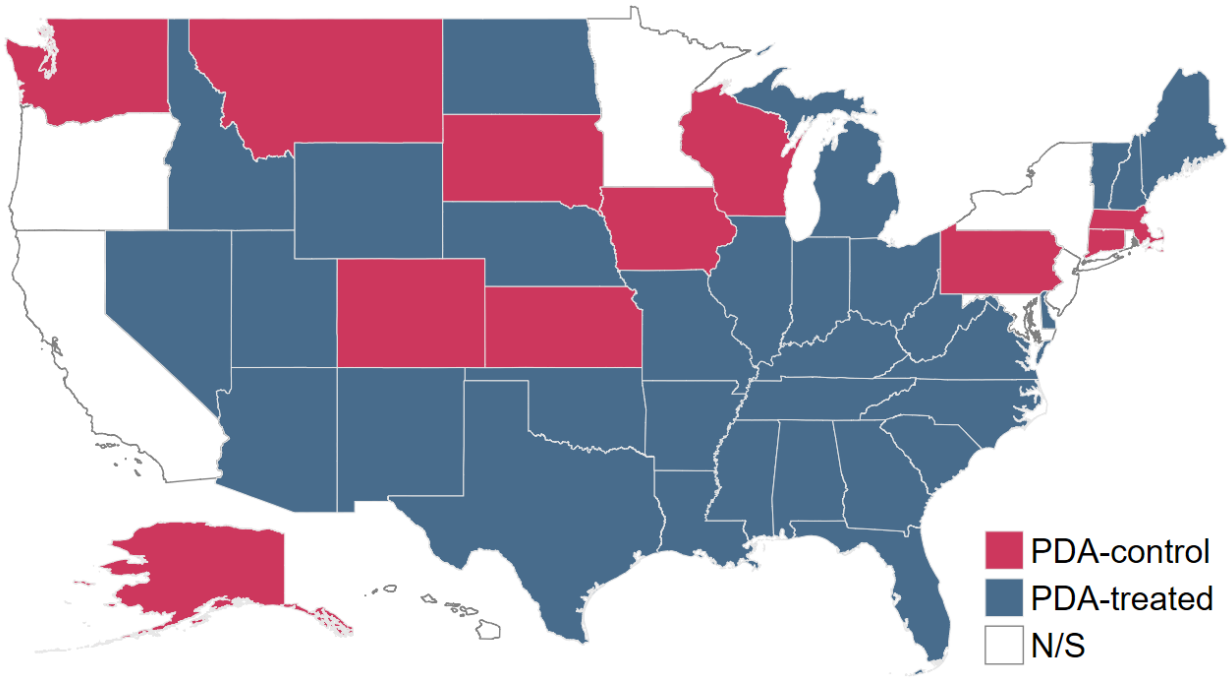
Notes: The sample is restricted to respondents age 18 to 60 in PDA-treated and PDA-control states between 1973 and 1984, from the CPS ASEC. Estimates are computed using ASEC individual sample weights. Employed is a dummy equal to 1 if the respondent is employed at some point in the year. Hired is a dummy equal to 1 if the respondent was not employed in the previous year and is employed in the previous week, and 0 if not employed in the previous week. PF stands for predicted fertility, which is computed as detailed in section A.6. Bars display the estimated coefficients on the interactions between the *PDA-treated* and the *Post-PDA* indicators and, separately, a dummy for having below-median PF (green) and above-median PF (orange). The regression includes all double and triple interactions between the *PDA-treated* and *Post-PDA* indicators and the dummy for above-median PF, and this dummy alone. 95% confidence intervals are based on heteroskedasticity robust standard errors.

Figure A8: DID Estimates for Employment, Robustness to Set of States



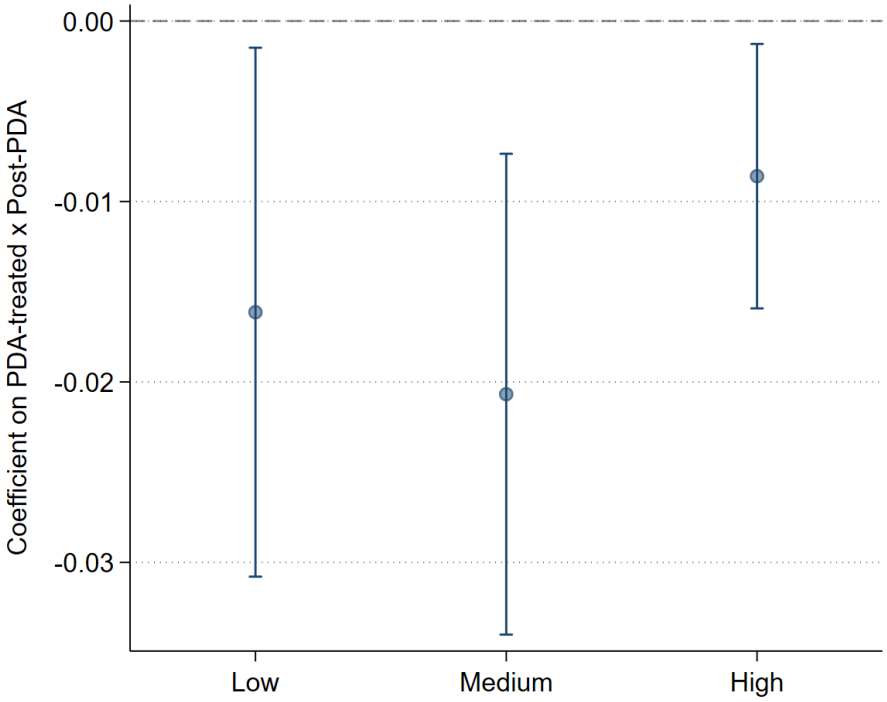
Notes: This plot displays the DID coefficient on the interaction between the *PDA-treated* and *Post-PDA* dummies from specification (7), estimated dropping from the sample observations from one state group at a time. The omitted state group is indicated on the x-axis. The main sample is taken from the CPS ASEC, in panel (a), and from the PSID, in panel (b). Estimates are computed using individual-level survey weights. 95% confidence intervals are based on heteroskedasticity robust standard errors.

Figure A9: PDA-treated and PDA-control States, PSID All Possible States



Notes: This map displays the set of PDA-treated and PDA-control states used in the robustness exercise using all possible states from PSID data.

Figure A10: Effect on Employment and Wages by Tercile of Industry-level Health Insurance Coverage, CPS ASEC data



Notes: The sample is restricted to women of age 18 to 35 in PDA-treated and PDA-control states between 1973 and 1984, from the CPS ASEC. Estimates are computed using ASEC individual sample weights. Employed is a dummy equal to 1 if the respondent is employed at some point in the year. Industries are grouped in terciles based on their share of workers covered by own employer-provided health insurance, which is measured from the 1976 SIE. 95% confidence intervals are based on heteroskedasticity robust standard errors.

A.9 Additional Tables

Table A2: Employment Protection for Pregnancy Legislative Changes Prior to 1978

State	Enactment Year	Mode of Passage	Policy Mandate
Alaska	1975	Legislature	Equal Treatment
Colorado	1972	Administrative	Equal Treatment, Unpaid Leave
Connecticut	1973	Legislature	Equal Treatment, Unpaid Leave
District of Columbia	1977	Legislature	Equal Treatment
Iowa	1975	State court	Equal treatment
Kansas	1972	Administrative	Equal Treatment, Unpaid Leave
Maryland	1977	Legislature	Equal treatment
Massachusetts	1972	Legislature	Unpaid Leave
Minnesota	1977	Legislature	Equal treatment
Montana	1975	Legislature	Equal Treatment, Unpaid Leave
New York	1974	State court	Equal treatment
Oregon	1977	Legislature	Equal treatment
Pennsylvania	1973	State court	Equal treatment
South Dakota	1975	Administrative	Equal treatment
Washington	1972	Administrative	Equal Treatment, Unpaid Leave
Wisconsin	1975	State court	Equal Treatment

Notes: Details about state policies and sources are reported in sections 3 and A.1. Mode of passage lists the institutional channel through which the policy was enacted: state legislature, administrative agency, or state supreme court decision. States not listed did not enact a policy comparable or more protective than the PDA before 1978. California passed an unpaid leave mandate in 1978, but it is not included in the sample since it passed an amendment to cover pregnancies via STDI in 1976. Michigan passed an equal treatment mandate in 1978 and it is thus used as a PDA-treated state. New York is not used as a PDA-treated state since it also passed an amendment to cover pregnancies via STDI in 1977.

Table A3: Difference-in-Differences Estimates, Basic Monthly CPS data

	Employed	Hired
	(1)	(2)
PDA-treated x Post-PDA	-0.030	-0.005
	(0.003)	(0.004)
Observations	975,402	247,408
Pre-PDA Mean	0.543	0.0852
R-squared	0.077	0.022
State FE	✓	✓
Year-Month FE	✓	✓

Notes: The sample is restricted to women of age 18 to 35 in PDA-treated and PDA-control states between 1976 and 1984, from the basic monthly CPS. Employed is a dummy equal to 1 if the respondent was employed in the previous week. Hired is a dummy equal to 1 if the respondent was not employed in the previous month and is employed in the previous week, and 0 if still not employed in the previous week. Column (2) is therefore restricted to respondents who can be linked to the previous month's CPS sample. In 1977, Hired can be constructed only for March, April, and June because of known CPS linking limitations (Flood et al., 2022). Estimates are computed using CPS individual sample weights. 95% confidence intervals are based on heteroskedasticity robust standard errors.

Table A4: Difference-in-Differences Estimates for Women 40-60 and Men 18-60

	Women 40-60		Men 18-60	
	Employed	Hourly Wage	Employed	Hourly Wage
	(1)	(2)	(3)	(4)
PDA-treated x Post-PDA	0.002	0.001	-0.004	-0.069
	(0.007)	(0.020)	(0.003)	(0.011)
Observations	109,342	51,691	204,212	142,949
Pre-PDA Mean	0.591	3.833	0.914	5.872
R-squared	0.075	0.169	0.061	0.276
State FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓

Notes: The sample is restricted to respondents with gender and age as per the column headers, in PDA-treated and PDA-control states between 1973 and 1984, from the CPS ASEC. Estimates are computed using ASEC individual sample weights and include the fixed effects indicated in the table. Standard errors are robust to heteroskedasticity. Employed is a dummy equal to 1 if the respondent is employed at some point in the year. Wage income is measured in US dollars and is pre-tax. Hourly wage income is constructed using annual pre-tax wage income, number of weeks employed and usual weekly working hours.

Table A5: Difference-in-Differences Estimates on State-level Employment and GDP

	Employees per-resident	Establishments per 1000 residents	Avg. Employees per Establishment	Log(GDP per capita)
	(1)	(2)	(3)	(4)
PDA-treated x Post-PDA	-0.019	-0.523	-0.284	-0.025
	(0.005)	(0.145)	(0.159)	(0.017)
Observations	179	179	179	216
Pre-PDA Mean	0.272	19.23	14.36	6750
R-squared	0.963	0.980	0.970	0.980
State FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓

Notes: The sample is restricted to PDA-treated and PDA-control states between 1973 and 1984. Outcomes in columns (1) to (3) are based on data from the CBP. CBP data cover aggregate employment and establishments for a subset of the workforce. Excluded categories are self-employed, private-household employees, railroad employees, agricultural-production employees, and most government employees. The outcome in column (4) is based on state GDP yearly estimates from BEA. Outcomes in columns (1), (2), and (4) are scaled by state population using intercensal population estimates from the Census. Estimates are computed including the fixed effects indicated in the table. Standard errors are robust to heteroskedasticity.

Table A6: Triple Difference-in-Differences Estimates

	(1) Employed	(2) Hourly Wage
PDA-treated x Post-PDA x W 18-35	-0.043	0.034
	(0.008)	(0.017)
Observations	409,825	252,634
Pre-PDA Mean	0.686	3.549
R-squared	0.132	0.294
State FE	✓	✓
Year FE	✓	✓
State-Year FE	✓	✓

Notes: The sample is restricted to observations in PDA-treated and PDA-control states between 1973 and 1984, from the CPS ASEC. The sample consists of three groups: women age 18 to 35, women age 40 to 60 and men age 18 to 60. Estimates are computed using individual sample weights and include the fixed effects indicated in the table. Standard errors are robust to heteroskedasticity. The specification is described in Appendix A.5 and it includes all double and triple interactions between the dummies for *PDA-treated* state, *Post-PDA* year, and being a woman age 18 to 35, along with this last dummy included independently. Employed is a dummy equal to 1 if the respondent is employed at some point in the year. The hourly wage is expressed as a z-score. The hourly wage is constructed using annual pre-tax wage income, number of weeks employed and usual weekly working hours.

Table A7: Robustness to using only Massachusetts as PDA-control state

Source	CPS ASEC			PSID			Basic Monthly CPS	
	(1) Employed	(2) Hired	(3) Hourly Wage	(4) Employed	(5) Hired	(6) Hourly Wage	(7) Employed	(8) Hired
PDA-treated x Post-PDA	-0.062	0.005	0.010	-0.084	-0.182	-0.202	-0.025	-0.011
	(0.011)	(0.015)	(0.032)	(0.035)	(0.081)	(0.126)	(0.005)	(0.006)
Observations	100,427	29,571	59,958	17,060	6,122	10,876	849,795	214,891
Pre-PDA Mean	0.693	0.0914	3.508	0.675	0.285	3.835	0.547	0.0874
R-squared	0.069	0.014	0.152	0.084	0.054	0.235	0.077	0.021
State FE	✓	✓	✓	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓	✓	✓	✓

Notes: The sample is restricted to women age 18 to 35 in PDA-treated states and Massachusetts between 1973 and 1984, from the data described in the column headers. Estimates are computed using individual sample weights and include the fixed effects indicated in the table. Time fixed effects are year fixed effects for the CPS ASEC and the PSID, and year-month fixed effects for the Basic Monthly CPS. Standard errors are robust to heteroskedasticity. Employed is a dummy equal to 1 if the respondent is employed at some point in the year, for the CPS ASEC and the PSID, and currently employed in the week preceding the survey for the Basic Monthly CPS. Hired is a dummy equal to 1 if the respondent was not employed in the previous year (month for the Monthly CPS) and is employed in the current one, and 0 if still not employed. The hourly wage is expressed as a z-score. The hourly wage is constructed using annual pre-tax wage income, number of weeks employed and usual weekly working hours.

Table A8: Difference-in-Differences Estimates on Additional Outcomes

	Part-time (1)	# Weeks Worked (2)	# Weekly Hours (3)	Currently Employed (4)	Completed High School (5)	Completed Some College (6)	Currently Married (7)
PDA-treated x Post-PDA	0.016	-0.157	-0.749	-0.050	-0.002	-0.024	-0.003
	(0.008)	(0.326)	(0.226)	(0.007)	(0.006)	(0.007)	(0.007)
Observations	82,283	70,749	70,699	115,794	115,794	115,794	115,794
Pre-PDA Mean	0.254	38.24	35.33	0.519	0.764	0.290	0.632
R-squared	0.034	0.069	0.034	0.074	0.042	0.055	0.227
State FE	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓

Notes: The sample is restricted to female respondents age 18 to 35 in PDA-treated and PDA-control states between 1973 and 1984, from the CPS ASEC. Estimates are computed using ASEC individual sample weights and include the fixed effects indicated in the table. Standard errors are robust to heteroskedasticity. The outcomes of each column are defined as: (1) a dummy equal to 1 if the respondent is employed part-time and 0 if full-time, missing otherwise; (2) number of weeks employed in the year, conditional on ever being employed in that year; (3) number of usual weekly hours, conditional on ever being employed in the year; (4) a dummy equal to 1 if employed in the week preceding the March CPS survey; (5) a dummy equal to 1 if completed high school education and 0 otherwise; (6) a dummy equal to 1 if completed some college school education and 0 otherwise; (7) a dummy equal to 1 if currently married and 0 otherwise.

Table A9: Heterogeneous Effects by Individual Characteristics

	(1) Employed	(2) Hourly Wage	(3) Employed	(4) Hourly Wage	(5) Employed	(6) Hourly Wage
PDA-treated x Post-PDA	-0.038	-0.016	-0.038	-0.012	-0.051	-0.011
	(0.009)	(0.022)	(0.007)	(0.019)	(0.009)	(0.020)
PDA-treated x Post-PDA x Married	-0.015	-0.023				
	(0.013)	(0.036)				
PDA-treated x Post-PDA x Black			-0.022	0.038		
			(0.028)	(0.070)		
PDA-treated x Post-PDA x College Educ.					0.022	-0.001
					(0.014)	(0.039)
Observations	115,217	68,374	115,217	68,374	115,217	68,374
R-squared	0.074	0.157	0.074	0.157	0.073	0.158
State FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓

Notes: The sample is restricted to women of age 18 to 35 in PDA-treated and PDA-control states between 1973 and 1984, from the CPS ASEC. Estimates are computed using ASEC individual sample weights. Estimates are computed using ASEC individual sample weights and include the fixed effects indicated in the table. Standard errors are robust to heteroskedasticity. Employed is a dummy equal to 1 if the respondent is employed at some point in the year. Hired is a dummy equal to 1 if the respondent was not employed in the previous year and is employed in the previous week, and 0 if not employed in the previous week. Fired is a dummy equal to 1 if an individual absent from work in the previous week is absent due to a layoff and 0 otherwise. Wage income is measured in US dollars and is pre-tax. Hourly wage income is constructed using annual pre-tax wage income, number of weeks employed and usual weekly working hours. Estimates are computed estimating equation (7) restricting the sample to the set of individuals described in the figures' labels.

Table A10: Robustness to Larger Set of States and to Using Childhood State

	Larger Set of States		Childhood State		Larger Set, Childhood State	
	(1) Employed	(2) Hourly Wage	(3) Employed	(4) Hourly Wage	(5) Employed	(6) Hourly Wage
PDA-treated x Post-PDA	-0.067	0.019	-0.025	0.038	-0.030	-0.014
	(0.019)	(0.044)	(0.022)	(0.060)	(0.019)	(0.044)
Observations	23,648	14,832	18,960	12,496	22,602	14,882
Pre-PDA Mean	0.649	4.168	0.663	3.925	0.654	4.236
R-squared	0.094	0.213	0.087	0.240	0.095	0.212
State FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓

Notes: The sample is restricted to women 18 to 35 in PDA-treated and PDA-control states between 1973 and 1984, from PSID data. The set of PDA-treated and PDA-control states in columns (1)-(2) and (5)-(6) is illustrated in Figure A9. Columns (3) and (4) use the baseline set of PDA-treated and PDA-control states, shown in Figure 1, but assign respondents treatment status based on their childhood state of residence instead of their current state of residence. Estimates are computed using individual sample weights and include the fixed effects indicated in the table. Standard errors are robust to heteroskedasticity. Employed is a dummy equal to 1 if the respondent is employed at some point in the year. The hourly wage is expressed as a z-score. The hourly wage is constructed using annual pre-tax wage income, number of weeks employed and usual weekly working hours.

Table A11: Difference-in-Differences High *versus* Low Predicted Fertility, 1960-1980 Census

Outcome:	Employed		
Source:	PDA-treated States (1)	PDA-control States (2)	All States (3)
Post-PDA x High PF	-0.025	-0.044	-0.035
	(0.002)	(0.004)	(0.002)
High PF	0.107	0.133	0.119
	(0.002)	(0.004)	(0.002)
Observations	1,169,993	341,636	2,141,577
Pre-PDA Mean	0.554	0.568	0.563
R-squared	0.113	0.147	0.122
State FE	✓	✓	✓
Year FE	✓	✓	✓

Notes: The sample is restricted to women age 18 to 35 from samples of the Censuses of 1960, 1970, and 1980, from the set of states described in the column headers. Estimates are computed using individual sample weights and include the fixed effects indicated in the table. Standard errors are robust to heteroskedasticity. Employed is a dummy equal to 1 if the respondent is employed at some point in the year. PF stands for predicted fertility, which is computed as detailed in section A.6.

Table A12: Robustness to Inclusion of Contemporaneous Policy Controls

Outcome:	Employed					
Source:	CPS ASEC			PSID		
	(1)	(2)	(3)	(4)	(5)	(6)
PDA-treated x Post-PDA	-0.043	-0.038	-0.044	-0.075	-0.057	-0.070
	(0.007)	(0.008)	(0.007)	(0.024)	(0.025)	(0.023)
Observations	109,612	115,217	115,217	18,484	19,197	19,197
Pre-PDA Mean	0.686	0.686	0.686	0.660	0.660	0.660
R-squared	0.074	0.073	0.073	0.089	0.087	0.087
State FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Head Start	✓			✓		
Job Protection		✓			✓	
UDL			✓			✓

Notes: The sample is restricted to women age 18 to 35 in PDA-treated and PDA-control states between 1973 and 1984, from the CPS ASEC, in columns (1) to (3), and from the PSID, in columns (4) to (6). Estimates are computed using ASEC and PSID individual sample weights. Employed is a dummy equal to 1 if the respondent is employed at some point in the year. Estimates are obtained from equation (7), augmented with the inclusion of the controls for contemporaneous policy listed at the bottom of the table. Contemporaneous policy controls are based on the data sources described in section 6.4.

Table A13: DID Estimates, Cluster-Robust Inference

Outcome:	Employed		
Source:	CPS ASEC	PSID	Basic Monthly CPS
	(1)	(2)	(3)
PDA-treated x Post-PDA	-0.045	-0.069	-0.030
	(0.007)	(0.022)	(0.003)
Regression-based p-values			
Heteroskedasticity robust	< 0.0001	0.0022	< 0.0001
Clustered (conventional)	0.0010	0.0054	0.0002
Bell and McCaffrey (2002)	0.0545	0.0885	0.0381
Pustejovsky and Tipton (2018)	0.0567	0.0889	0.0377
Placebo inference from Hagemann (2019)			
PDA-treated x Post-PDA	-0.0508	-0.0610	-0.0415
p-value	0.0091	0.0613	0.0098
Observations	115,217	19,197	975,402
Clusters	12	18	18
Treated Clusters	9	15	15
Pre-PDA Mean	0.6859	0.6598	0.5431
R-squared	0.073	0.087	0.077
State FE	✓	✓	✓
Time FE	✓	✓	✓

Notes: The sample is restricted to women age 18 to 35 in PDA-treated and PDA-control states between 1973 and 1984, from the data described in the column headers. Estimates are computed using individual sample weights and include the fixed effects indicated in the table. Time fixed effects are year fixed effects for the CPS ASEC and the PSID, and year-month fixed effects for the Basic Monthly CPS. Standard errors are robust to heteroskedasticity. Employed is a dummy equal to 1 if the respondent is employed at some point in the year, for the CPS ASEC and the PSID, and currently employed in the week preceding the survey for the Basic Monthly CPS.